

Masked Autoencoders Are Scalable Vision Learners

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Abstract

This paper shows that masked autoencoders (MAE) are scalable self-supervised learners for computer vision. Our MAE approach is simple: we mask random patches of the input image and reconstruct the missing pixels. It is based on two core designs. First, we develop an asymmetric encoder-decoder architecture, with an encoder that operates only on the visible subset of patches (without mask tokens), along with a lightweight decoder that reconstructs the original image from the latent representation and mask tokens. Second, we find that masking a high proportion of the input image, e.g., 75%, yields a nontrivial and meaningful self-supervisory task. Coupling these two designs enables us to train large models efficiently and effectively: we accelerate training (by 3× or more) and improve accuracy. Our scalable approach allows for learning high-capacity models that generalize well: e.g., a vanilla ViT-Huge model achieves the best accuracy (87.8%) among methods that use only ImageNet-1K data. Transfer performance in downstream tasks outperforms supervised pre-training and shows promising scaling behavior.

1. Introduction

Deep learning has witnessed an explosion of architectures of continuously growing capability and capacity [33, 25, 57]. Aided by the rapid gains in hardware, models today can easily overfit one million images [13] and begin to demand hundreds of millions of—often publicly inaccessible—labeled images [16].

This appetite for data has been successfully addressed in natural language processing (NLP) by self-supervised pre-training. The solutions, based on autoregressive language modeling in GPT [47, 48, 4] and masked autoencoding in BERT [14], are conceptually simple: they remove a portion of the data and learn to predict the removed content. These methods now enable training of generalizable NLP models containing over one hundred billion parameters [4].

The idea of masked autoencoders, a form of more general denoising autoencoders [58], is natural and applicable in computer vision as well. Indeed, closely related research

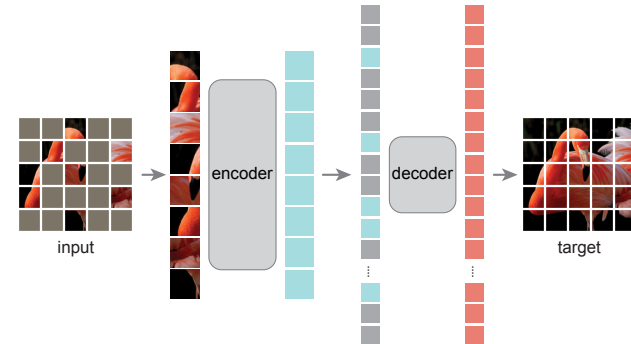


Figure 1. **Our MAE architecture.** During pre-training, a large random subset of image patches (e.g., 75%) is masked out. The encoder is applied to the small subset of visible patches. Mask tokens are introduced after the encoder, and the full set of encoded patches and mask tokens is processed by a small decoder that reconstructs the original image in pixels. After pre-training, the decoder is discarded and the encoder is applied to uncorrupted images (full sets of patches) for recognition tasks.

in vision [59, 46] preceded BERT. However, despite significant interest in this idea following the success of BERT, progress of autoencoding methods in vision lags behind NLP. We ask: *what makes masked autoencoding different between vision and language?* We attempt to answer this question from the following perspectives:

(i) Until recently, architectures were different. In vision, convolutional networks [34] were dominant over the last decade [33]. Convolutions typically operate on regular grids and it is not straightforward to integrate ‘indicators’ such as mask tokens [14] or positional embeddings [57] into convolutional networks. This architectural gap, however, has been addressed with the introduction of Vision Transformers (ViT) [16] and should no longer present an obstacle.

(ii) Information density is different between language and vision. Languages are human-generated signals that are highly semantic and information-dense. When training a model to predict only a few missing words per sentence, this task appears to induce sophisticated language understanding. Images, on the contrary, are natural signals with heavy spatial redundancy—e.g., a missing patch can be recovered from neighboring patches with little high-level un-

掩码自编码器是可扩展的视觉学习器

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摘要

本文表明，掩码自编码器 (MAE) 是可扩展的计算机视觉自监督学习器。MAE 方法很简单：随机遮蔽输入图像的图块，并重建缺失的像素。它基于两个核心设计。首先，我们提出非对称的编码器-解码器架构：编码器只处理可见图块子集（不使用掩码令牌），轻量解码器则根据潜在表征与掩码令牌重建原图。其次，我们发现遮蔽很大比例的输入图像（如 75%），能产生不平凡且有意义的自监督任务。结合这两个设计，可以高效且有效地训练大型模型：训练加速 3× 或更多，同时提升准确率。这种可扩展的方法能够学习泛化良好的大容量模型，例如普通 ViT-Huge 在仅使用 ImageNet-1K 数据的方法中达到最高准确率 (87.8%)。在下游任务中的迁移性能优于监督预训练，并展现出有前景的规模扩展行为。

1. 引言

深度学习见证了能力和容量不断增长的架构迅速涌现[33, 25, 57]。借助硬件快速进步，当前模型很容易在一百万张图像上过拟合 [13]，开始需要数亿张往往不公开的标注图像 [16]。

NLP 已通过自监督预训练成功满足这种数据需求。GPT 的自回归语言建模 [47, 48, 4] 和 BERT 的掩码自编码 [14]，在概念上很简单：移除部分数据，并学习预测被移除的内容。这些方法目前已支持训练拥有超过千亿参数、能泛化的 NLP 模型 [4]。

掩码自编码器作为更广义去噪自编码器 [58] 的一种形式，在计算机视觉中也是自然且适用的思路。事实上，紧密相关的研究

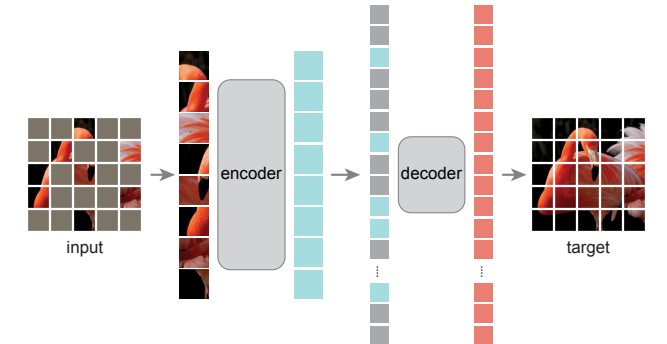


图 1. MAE 架构。预训练时，随机遮蔽大比例图块（如 75%）。编码器只处理少量可见图块。编码器之后才引入掩码令牌，编码图块与掩码令牌的完整集合由小型解码器处理，在像素空间重建原图。预训练后丢弃解码器，编码器处理完整、未破坏的图像（全部图块）以执行识别任务。

视觉中的相关研究 [59, 46] 甚至早于 BERT。但尽管 BERT 的成功引起了对这一思路的浓厚兴趣，视觉自编码方法的进展仍落后于 NLP。我们追问：视觉和语言中的掩码自编码为何不同？下面从几个角度尝试回答。

(i) 直到最近，二者架构仍不同。在视觉中，卷积网络 [34] 在过去十年 [33] 占据主导。卷积通常作用于规则网格，将掩码令牌 [14] 或位置嵌入 [57] 等“指示信息”融入卷积网络并不直接。不过，Vision Transformer (ViT) [16] 的引入已填补这一架构差距，因此不应再成为障碍。

(ii) 语言与视觉的信息密度不同。语言是人类生成、高度语义化且信息密集的信号。训练模型预测每句中少数缺失词，似乎即可促成复杂的语言理解。相反，图像是具有大量空间冗余的自然信号，例如无须深入理解局部、物体和场景，就可通过相邻图块恢复缺失图块，

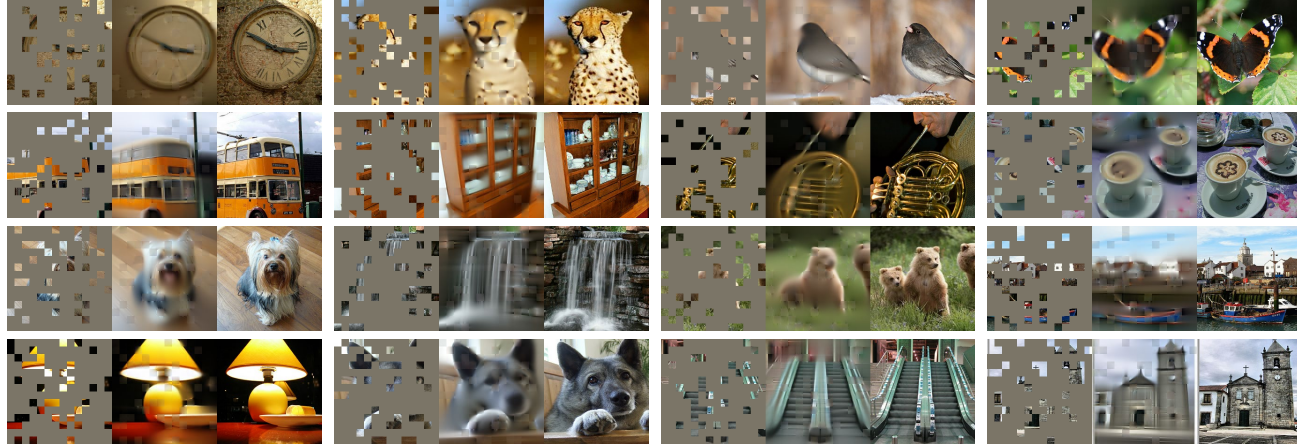


Figure 2. Example results on ImageNet *validation* images. For each triplet, we show the masked image (left), our MAE reconstruction[†] (middle), and the ground-truth (right). The masking ratio is 80%, leaving only 39 out of 196 patches. More examples are in the appendix.
[†]As no loss is computed on visible patches, the model output on visible patches is qualitatively worse. One can simply overlay the output with the visible patches to improve visual quality. We intentionally opt not to do this, so we can more comprehensively demonstrate the method’s behavior.



Figure 3. Example results on COCO validation images, using an MAE trained on ImageNet (the same model weights as in Figure 2). Observe the reconstructions on the two right-most examples, which, although different from the ground truth, are semantically plausible.

understanding of parts, objects, and scenes. To overcome this difference and encourage learning useful features, we show that a simple strategy works well in computer vision: masking a *very high* portion of random patches. This strategy largely reduces redundancy and creates a challenging self-supervisory task that requires holistic understanding beyond low-level image statistics. To get a qualitative sense of our reconstruction task, see Figures 2 – 4.

(iii) The autoencoder’s *decoder*, which maps the latent representation back to the input, plays a different role between reconstructing text and images. In vision, the decoder reconstructs *pixels*, hence its output is of a lower semantic level than common recognition tasks. This is in contrast to language, where the decoder predicts missing *words* that contain rich semantic information. While in BERT the decoder can be trivial (an MLP) [14], we found that for images, the decoder design plays a key role in determining the semantic level of the learned latent representations.

Driven by this analysis, we present a simple, effective, and scalable form of a masked autoencoder (MAE) for visual representation learning. Our MAE masks random patches from the input image and reconstructs the missing patches in the pixel space. It has an *asymmetric* encoder-decoder design. Our encoder operates only on the visible subset of patches (without mask tokens), and our decoder is

lightweight and reconstructs the input from the latent representation along with mask tokens (Figure 1). Shifting the mask tokens to the small decoder in our asymmetric encoder-decoder results in a large reduction in computation. Under this design, a very high masking ratio (*e.g.*, 75%) can achieve a win-win scenario: it optimizes accuracy while allowing the encoder to process only a small portion (*e.g.*, 25%) of patches. This can reduce overall pre-training time by 3× or more and likewise reduce memory consumption, enabling us to easily scale our MAE to large models.

Our MAE learns very high-capacity models that generalize well. With MAE pre-training, we can train data-hungry models like ViT-Large/-Huge [16] on ImageNet-1K with improved generalization performance. With a vanilla ViT-Huge model, we achieve 87.8% accuracy when fine-tuned on ImageNet-1K. This outperforms all previous results that use only ImageNet-1K data. We also evaluate transfer learning on object detection, instance segmentation, and semantic segmentation. In these tasks, our pre-training achieves better results than its supervised pre-training counterparts, and more importantly, we observe significant gains by scaling up models. These observations are aligned with those witnessed in self-supervised pre-training in NLP [14, 47, 48, 4] and we hope that they will enable our field to explore a similar trajectory.

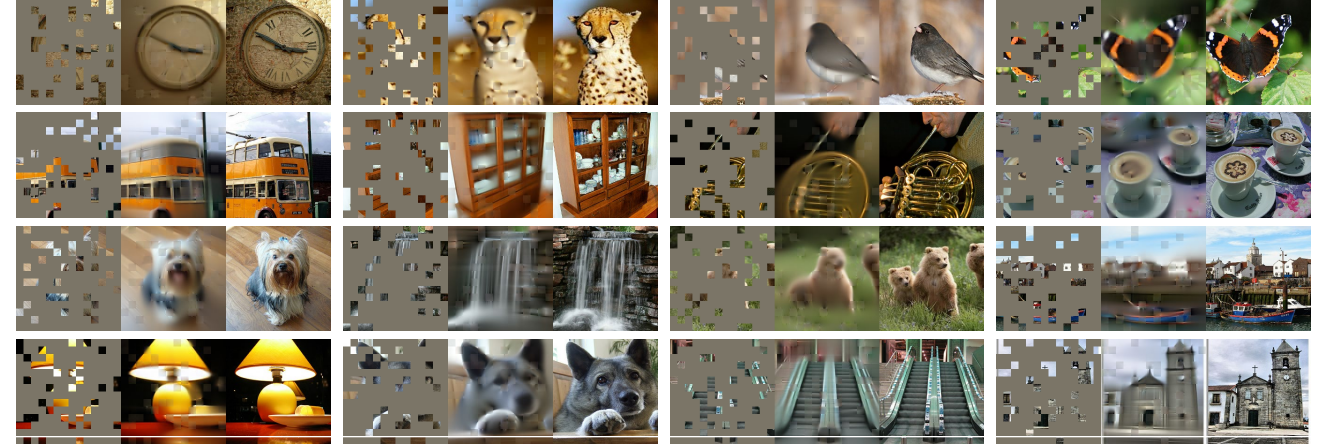


图 2. ImageNet 验证图像上的示例结果。每组三幅图依次为遮蔽图像（左）、MAE 重建[†]（中）和真实图像（右）。掩码比例为 80%，196 个图块仅保留 39 个。附录提供更多示例。
[†]由于可见图块不计算损失，模型在这些图块上的输出质量较差。直接将可见图块覆盖到输出上即可改善视觉质量。我们特意没有这样做，以更全面地展示方法的行为。



图 3. COCO 验证图像上的示例，使用在 ImageNet 上训练的 MAE（权重与图 2 相同）。观察最右侧两个示例的重建：虽然与真实图像不同，但语义上合理。

而不必掌握这些高层知识。为克服这种差异并鼓励学习有用特征，我们展示了视觉中有效的简单策略：随机遮蔽很高比例的图块。该策略大幅减少冗余，构成具有挑战性的自监督任务，要求超越低层图像统计的整体理解。图 2–4 可直观展示重建任务。

(iii) 自编码器的解码器将潜在表征映射回输入，但在文本重建和图像重建中承担不同角色。在视觉中，解码器重建像素，因此输出的语义层次低于常见识别任务。语言中则不同，解码器预测的缺失词语包含丰富语义。在 BERT 中解码器可以非常简单（一个 MLP）[14]，但我们发现，在图像中，解码器设计对决定潜在表征的语义层次至关重要。

根据上述分析，我们提出简单、有效且可扩展的掩码自编码器（MAE），用于视觉表征学习。MAE 随机遮蔽输入图块，在像素空间重建缺失图块。它采用非对称编码器–解码器设计：编码器仅处理可见图块子集（不使用掩码令牌），解码器则是

轻量的，并结合潜在表征和掩码令牌重建输入（图 1）。在非对称编码器–解码器中，将掩码令牌移至小型解码器，可大幅减少计算。采用此设计后，很高的掩码比例（如 75%）带来双赢：既优化准确率，又让编码器只需处理一小部分（如 25%）图块。这可使整体预训练提速 3× 或更多，并降低内存消耗，使 MAE 易于扩展到大型模型。

我们的 MAE 能学习具有很高容量、且泛化良好的模型。通过 MAE 预训练，可以在 ImageNet-1K 上训练 ViT-Large/-Huge [16] 这类数据需求很大的模型，并改善其泛化性能。使用普通的 ViT-Huge 模型，在 ImageNet-1K 上微调后达到 87.8% 的准确率，超过此前所有仅使用 ImageNet-1K 数据的结果。我们还评估了目标检测、实例分割和语义分割中的迁移学习。在这些任务中，我们的预训练优于相应的监督预训练；更重要的是，扩大模型规模带来了显著收益。这些观察与 NLP 自监督预训练中的发现[14, 47, 48, 4]一致，我们希望它们能推动视觉领域探索类似的发展路径。

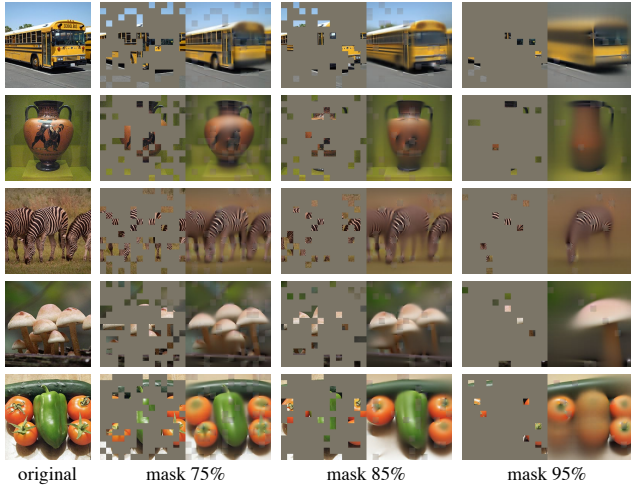


Figure 4. Reconstructions of ImageNet *validation* images using an MAE pre-trained with a masking ratio of 75% but applied on inputs with higher masking ratios. The predictions differ plausibly from the original images, showing that the method can generalize.

2. Related Work

Masked language modeling and its autoregressive counterparts, *e.g.*, BERT [14] and GPT [47, 48, 4], are highly successful methods for pre-training in NLP. These methods hold out a portion of the input sequence and train models to predict the missing content. These methods have been shown to scale excellently [4] and a large abundance of evidence indicates that these pre-trained representations generalize well to various downstream tasks.

Autoencoding is a classical method for learning representations. It has an encoder that maps an input to a latent representation and a decoder that reconstructs the input. For example, PCA and k-means are autoencoders [29]. Denoising autoencoders (DAE) [58] are a class of autoencoders that corrupt an input signal and learn to reconstruct the original, uncorrupted signal. A series of methods can be thought of as a generalized DAE under different corruptions, *e.g.*, masking pixels [59, 46, 6] or removing color channels [70]. Our MAE is a form of denoising autoencoding, but different from the classical DAE in numerous ways.

Masked image encoding methods learn representations from images corrupted by masking. The pioneering work of [59] presents masking as a noise type in DAE. Context Encoder [46] inpaints large missing regions using convolutional networks. Motivated by the success in NLP, related recent methods [6, 16, 2] are based on Transformers [57]. iGPT [6] operates on sequences of pixels and predicts unknown pixels. The ViT paper [16] studies masked patch prediction for self-supervised learning. Most recently, BEiT [2] proposes to predict discrete tokens [44, 50].

Self-supervised learning approaches have seen significant interest in computer vision, often focusing on different pre-text tasks for pre-training [15, 61, 42, 70, 45, 17]. Recently, contrastive learning [3, 22] has been popular, *e.g.*, [62, 43, 23, 7], which models image similarity and dissimilarity (or only similarity [21, 8]) between two or more views. Contrastive and related methods strongly depend on data augmentation [7, 21, 8]. Autoencoding pursues a conceptually different direction, and it exhibits different behaviors as we will present.

3. Approach

Our masked autoencoder (MAE) is a simple autoencoding approach that reconstructs the original signal given its partial observation. Like all autoencoders, our approach has an encoder that maps the observed signal to a latent representation, and a decoder that reconstructs the original signal from the latent representation. Unlike classical autoencoders, we adopt an *asymmetric* design that allows the encoder to operate only on the partial, observed signal (without mask tokens) and a lightweight decoder that reconstructs the full signal from the latent representation and mask tokens. Figure 1 illustrates the idea, introduced next.

Masking. Following ViT [16], we divide an image into regular non-overlapping patches. Then we sample a subset of patches and mask (*i.e.*, remove) the remaining ones. Our sampling strategy is straightforward: we sample random patches without replacement, following a uniform distribution. We simply refer to this as “random sampling”.

Random sampling with a *high* masking ratio (*i.e.*, the ratio of removed patches) largely eliminates redundancy, thus creating a task that cannot be easily solved by extrapolation from visible neighboring patches (see Figures 2 – 4). The uniform distribution prevents a potential center bias (*i.e.*, more masked patches near the image center). Finally, the highly sparse input creates an opportunity for designing an efficient encoder, introduced next.

MAE encoder. Our encoder is a ViT [16] but applied only on *visible, unmasked patches*. Just as in a standard ViT, our encoder embeds patches by a linear projection with added positional embeddings, and then processes the resulting set via a series of Transformer blocks. However, our encoder only operates on a small subset (*e.g.*, 25%) of the full set. Masked patches are removed; no mask tokens are used. This allows us to train very large encoders with only a fraction of compute and memory. The full set is handled by a lightweight decoder, described next.

MAE decoder. The input to the MAE decoder is the full set of tokens consisting of (i) encoded visible patches, and (ii) mask tokens. See Figure 1. Each mask token [14] is a shared, learned vector that indicates the presence of a miss-

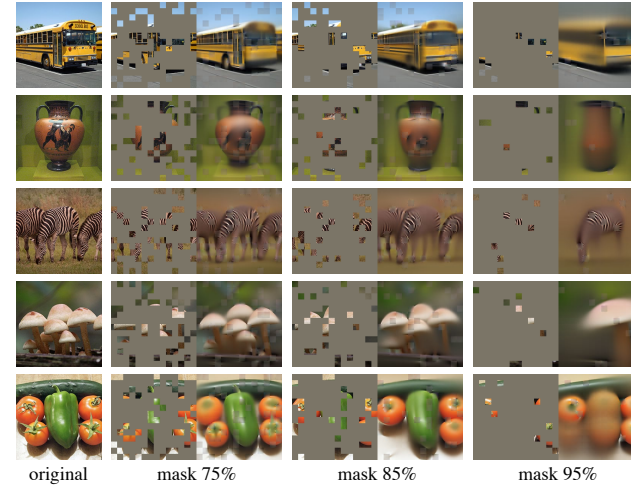


图 4. ImageNet 验证图像的重建。MAE 以 75% 掩码比例预训练，但应用于更高掩码比例输入。预测与原图虽不同，却仍合理，说明方法能够泛化。

2. 相关工作

掩码语言建模及其自回归对应方法（如 BERT [14]和 GPT [47, 48, 4],）是极其成功的 NLP 预训练方法。它们留出部分输入序列，训练模型预测缺失内容。这些方法具有很好的规模扩展能力 [4]，大量证据也表明，所得预训练表征能很好地泛化至各种下游任务。

自编码是一种经典的表征学习方法。编码器将输入映射到潜在表征，解码器重建输入。例如，PCA 和 k-means 都是自编码器 [29]。去噪自编码器（DAE）[58]是一类先破坏输入信号、再学习重建原始未受损信号的自编码器。一系列方法可视为不同破坏方式下的广义 DAE，例如遮蔽像素 [59, 46, 6]或移除颜色通道 [70]。MAE 属于去噪自编码，但在许多方面与传统 DAE 不同。

掩码图像编码方法通过被遮蔽破坏的图像学习表征。先驱工作 [59] 将遮蔽作为 DAE 中的一种噪声。Context Encoder [46] 使用卷积网络补全大面积缺失区域。受 NLP 成功启发，近期相关方法 [6, 16, 2] 基于 Transformer [57]。iGPT [6] 处理像素序列并预测未知像素。ViT 论文 [16] 研究用掩码图块预测进行自监督学习。最近，BEiT[2] 提出预测离散令牌 [44, 50]。

计算机视觉中的自监督学习受到广泛关注，通常围绕不同的预训练前置任务 [15, 61, 42, 70, 45, 17]展开。最近，对比学习 [3, 22] 十分流行，例如 [62, 43, 23, 7],，它对两个或多个视图之间的图像相似性与不相似性（或仅相似性 [21, 8]）建模。对比学习及相关方法强烈依赖数据增强 [7, 21, 8]。自编码探索的是概念上不同的方向，并表现出下文将展示的不同性质。

3. 方法

掩码自编码器（MAE）是一种简单的自编码方法，根据部分观测重建原始信号。与所有自编码器一样，它有将观测信号映射到潜在表征的编码器，以及从潜在表征重建原始信号的解码器。与传统自编码器不同，我们采用非对称设计：编码器只处理部分可见信号（不带掩码令牌），轻量解码器则根据潜在表征与掩码令牌重建完整信号。图 1 展示了这一思路，下面具体介绍。

遮蔽。遵循 ViT [16],，将图像划分为规则、不重叠的图块，然后抽取一个子集，遮蔽（即移除）剩余图块。抽样策略很直接：按均匀分布，无放回地随机抽取图块，我们简称为“随机抽样”。

高掩码比例（即移除图块的比例）的随机抽样可大幅消除冗余，形成难以仅从可见相邻图块外推解决的任务（图 2–4）。均匀分布避免潜在的中心偏置，即图像中心附近被遮蔽更多。最后，高度稀疏的输入为下文的高效编码器设计创造了机会。

MAE 编码器。编码器是 ViT [16]，但仅作用于可见、未遮蔽图块。与标准 ViT 一样，它通过线性投影和位置嵌入编码图块，再通过一系列 Transformer 块处理所得集合。但编码器只处理完整集合的一小部分（如 25%）。被遮蔽图块直接移除，不使用掩码令牌。因此只需部分计算和内存即可训练超大型编码器。完整集合交由下文介绍的轻量解码器处理。

MAE 解码器。输入为完整令牌集合，包括 (i) 编码后的可见图块与 (ii) 掩码令牌，见图 1。每个掩码令牌 [14] 都是一个共享的可学习向量，用于指示存在一个待预测的缺失

ing patch to be predicted. We add positional embeddings to all tokens in this full set; without this, mask tokens would have no information about their location in the image. The decoder has another series of Transformer blocks.

The MAE decoder is only used during pre-training to perform the image reconstruction task (only the encoder is used to produce image representations for recognition). Therefore, the decoder architecture can be flexibly designed in a manner that is *independent* of the encoder design. We experiment with very small decoders, narrower and shallower than the encoder. For example, our default decoder has $<10\%$ computation per token vs. the encoder. With this asymmetrical design, the full set of tokens are only processed by the lightweight decoder, which significantly reduces pre-training time.

Reconstruction target. Our MAE reconstructs the input by predicting the *pixel* values for each masked patch. Each element in the decoder’s output is a vector of pixel values representing a patch. The last layer of the decoder is a linear projection whose number of output channels equals the number of pixel values in a patch. The decoder’s output is reshaped to form a reconstructed image. Our loss function computes the mean squared error (MSE) between the reconstructed and original images in the pixel space. We compute the loss only on masked patches, similar to BERT [14].¹

We also study a variant whose reconstruction target is the normalized pixel values of each masked patch. Specifically, we compute the mean and standard deviation of all pixels in a patch and use them to normalize this patch. Using normalized pixels as the reconstruction target improves representation quality in our experiments.

Simple implementation. Our MAE pre-training can be implemented efficiently, and importantly, does not require any specialized sparse operations. First we generate a token for every input patch (by linear projection with an added positional embedding). Next we *randomly shuffle* the list of tokens and *remove* the last portion of the list, based on the masking ratio. This process produces a small subset of tokens for the encoder and is equivalent to sampling patches without replacement. After encoding, we append a list of mask tokens to the list of encoded patches, and *unshuffle* this full list (inverting the random shuffle operation) to align all tokens with their targets. The decoder is applied to this full list (with positional embeddings added). As noted, no sparse operations are needed. This simple implementation introduces negligible overhead as the shuffling and unshuffling operations are fast.

¹Computing the loss only on masked patches differs from traditional denoising autoencoders [58] that compute the loss on all pixels. This choice is purely result-driven: computing the loss on all pixels leads to a slight decrease in accuracy (e.g., $\sim 0.5\%$).

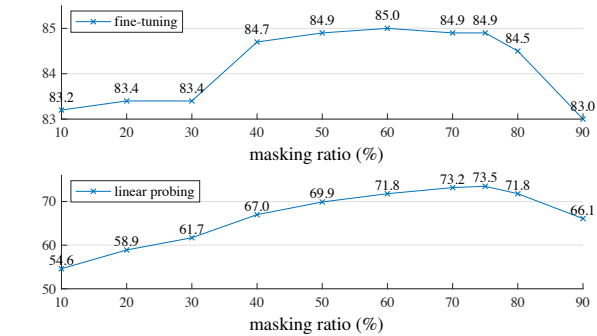


Figure 5. **Masking ratio.** A high masking ratio (75%) works well for both fine-tuning (top) and linear probing (bottom). The y-axes are ImageNet-1K validation accuracy (%) in all plots in this paper.

4. ImageNet Experiments

We do self-supervised pre-training on the ImageNet-1K (IN1K) [13] training set. Then we do supervised training to evaluate the representations with (i) end-to-end fine-tuning or (ii) linear probing. We report top-1 validation accuracy of a single 224×224 crop. Details are in Appendix A.1.

Baseline: ViT-Large. We use ViT-Large (ViT-L/16) [16] as the backbone in our ablation study. ViT-L is very big (an order of magnitude bigger than ResNet-50 [25]) and tends to overfit. The following is a comparison between ViT-L trained from scratch vs. fine-tuned from our baseline MAE:

scratch, original [16]	scratch, our impl.	baseline MAE
76.5	82.5	84.9

We note that it is nontrivial to train *supervised* ViT-L from scratch and a good recipe with strong regularization is needed (82.5%, see Appendix A.2). Even so, our MAE pre-training contributes a big improvement. Here fine-tuning is only for 50 epochs (vs. 200 from scratch), implying that the fine-tuning accuracy heavily depends on pre-training.

4.1. Main Properties

We ablate our MAE using the default settings in Table 1 (see caption). Several intriguing properties are observed.

Masking ratio. Figure 5 shows the influence of the masking ratio. The optimal ratios are surprisingly high. The ratio of 75% is good for both linear probing and fine-tuning. This behavior is in contrast with BERT [14], whose typical masking ratio is 15%. Our masking ratios are also much higher than those in related works [6, 16, 2] in computer vision (20% to 50%).

The model *infers* missing patches to produce different, yet plausible, outputs (Figure 4). It makes sense of the gestalt of objects and scenes, which cannot be simply completed by extending lines or textures. We hypothesize that this reasoning-like behavior is linked to the learning of useful representations.

Figure 5 also shows that linear probing and fine-tuning results follow *different* trends. For linear probing, the ac-

图块。我们给完整集中的所有令牌加入位置嵌入，否则掩码令牌无法获知自己在图像中的位置。解码器另包含一系列 Transformer 块。

MAE 解码器仅在预训练中用于图像重建（识别时只用编码器生成图像表征）。因此，解码器架构可独立于编码器灵活设计。我们实验了比编码器更窄、更浅的极小解码器。例如，默认解码器每个令牌的计算量小于编码器的 $<10\%$ 。借助这种非对称设计，完整令牌集合只由轻量解码器处理，从而显著减少预训练时间。

重建目标。MAE 通过预测每个被遮蔽图块的像素值来重建输入。解码器的每个输出元素都是一个表示图块的像素值向量。解码器最后一层为线性投影，输出通道数等于一个图块的像素值数量。将解码器输出重排可得到重建图像。损失函数计算重建图与原图在像素空间中的均方误差（MSE）。类似 BERT [14]，损失只在被遮蔽图块上计算。¹

我们还研究了以每个被遮蔽图块的归一化像素值为重建目标的变体。具体而言，计算图块内所有像素的均值和标准差，再用它们归一化该图块。在实验中，归一化像素目标改善了表征质量。

简单实现。MAE 预训练可以高效实现，而且不需要特殊的稀疏运算。首先，通过线性投影并添加位置嵌入，为每个输入图块生成令牌。随后随机打乱令牌列表，并按掩码比例移除列表末尾的一部分。这为编码器产生一个很小的令牌子集，等价于无放回抽样图块。编码完成后，向编码图块列表追加掩码令牌，并对整个列表恢复原序（随机打乱的逆操作），使所有令牌与其目标对齐。解码器处理这一

完整列表（添加位置嵌入后）。如前所述，无需稀疏运算。由于打乱与恢复顺序很快，这种简单实现仅带来可忽略的开销。

¹仅在被遮蔽图块上计算损失，不同于在全部像素上计算损失的传统去噪自编码器 [58]。这一选择完全由实验结果驱动：在全部像素上计算损失会使准确率略降（如 $\sim 0.5\%$ ）。

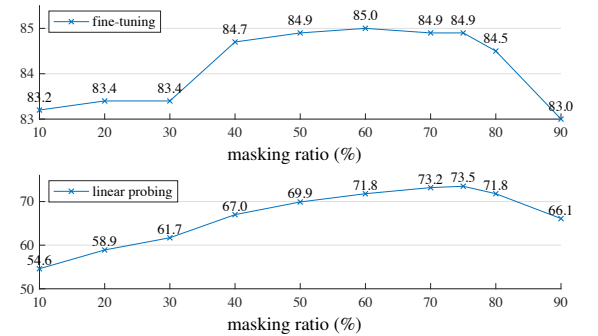


图 5. 掩码比例。较高比例（75%）对微调（上）和线性探测（下）均有效。本文所有图的纵轴均为 ImageNet-1K 验证准确率（%）。

4. ImageNet 实验

在 ImageNet-1K (IN1K) [13] 训练集上进行自监督预训练，再通过监督训练，用 (i) 端到端微调或 (ii) 线性探测评估表征。报告单个 224×224 裁剪的 top-1 验证准确率。详见附录 A.1。

基线：ViT-Large。消融使用 ViT-Large (ViT-L/16) [16] 作为骨干。ViT-L 很大（比 ResNet-50 [25] 大一个数量级），容易过拟合。下面比较从零训练 ViT-L 与从基线 MAE 微调的结果：

scratch, original [16]	scratch, our impl.	baseline MAE
76.5	82.5	84.9

需要指出，从零监督训练 ViT-L 并不容易，必须使用强正则化的良好方案（82.5%，见附录 A.2）。即便如此，MAE 预训练仍带来很大提升。这里微调仅 50 轮（从零训练则为 200 轮），说明微调准确率高度依赖预训练。

4.1. 主要性质

按表 1 默认设置（见表注）进行 MAE 消融，观察到若干有趣性质。掩码比例。图 5 展示其影响。最优比例出乎意料地高：75% 对线性探测和微调都很好。这不同于典型掩码比例为 15% 的 BERT [14]。我们的掩码比例也远高于视觉相关工作 [6, 16, 2] 的 20% 至 50%。

模型推断缺失图块，产生不同但合理的输出（图 4）。它理解物体和场景的整体形态，而不能仅靠延长线条或纹理完成任务。我们推测，这种类似推理的行为与有用表征的学习有关。

图 5 还说明，线性探测和微调遵循不同趋势。在线性探测中，

blocks	ft	lin
1	84.8	65.5
2	84.9	70.0
4	84.9	71.9
8	84.9	73.5
12	84.4	73.3

(a) **Decoder depth.** A deep decoder can improve linear probing accuracy.

dim	ft	lin
128	84.9	69.1
256	84.8	71.3
512	84.9	73.5
768	84.4	73.1
1024	84.3	73.1

(b) **Decoder width.** The decoder can be narrower than the encoder (1024-d).

case	ft	lin	FLOPs
encoder w/ [M]	84.2	59.6	3.3×
encoder w/o [M]	84.9	73.5	1×

(c) **Mask token.** An encoder without mask tokens is more accurate and faster (Table 2).

case	ft	lin
pixel (w/o norm)	84.9	73.5
pixel (w/ norm)	85.4	73.9
PCA	84.6	72.3
dVAE token	85.3	71.6

(d) **Reconstruction target.** Pixels as reconstruction targets are effective.

case	ft	lin
none	84.0	65.7
crop, fixed size	84.7	73.1
crop, rand size	84.9	73.5
crop + color jit	84.3	71.9

(e) **Data augmentation.** Our MAE works with minimal or no augmentation.

case	ratio	ft	lin
random	75	84.9	73.5
block	50	83.9	72.3
block	75	82.8	63.9
grid	75	84.0	66.0

(f) **Mask sampling.** Random sampling works the best. See Figure 6 for visualizations.

Table 1. **MAE ablation experiments** with ViT-L/16 on ImageNet-1K. We report fine-tuning (ft) and linear probing (lin) accuracy (%). If not specified, the default is: the decoder has depth 8 and width 512, the reconstruction target is unnormalized pixels, the data augmentation is random resized cropping, the masking ratio is 75%, and the pre-training length is 800 epochs. Default settings are marked in **gray**.

curacy increases steadily with the masking ratio until the sweet point: the accuracy gap is up to ~20% (54.6% vs. 73.5%). For fine-tuning, the results are less sensitive to the ratios, and a wide range of masking ratios (40–80%) work well. All fine-tuning results in Figure 5 are better than training from scratch (82.5%).

Decoder design. Our MAE decoder can be flexibly designed, as studied in Table 1a and 1b.

Table 1a varies the decoder depth (number of Transformer blocks). A sufficiently deep decoder is important for linear probing. This can be explained by the gap between a pixel reconstruction task and a recognition task: the last several layers in an autoencoder are more specialized for reconstruction, but are less relevant for recognition. A reasonably deep decoder can account for the reconstruction specialization, leaving the latent representations at a more abstract level. This design can yield up to 8% improvement in linear probing (Table 1a, ‘lin’). However, if fine-tuning is used, the last layers of the encoder can be tuned to adapt to the recognition task. The decoder depth is less influential for improving fine-tuning (Table 1a, ‘ft’).

Interestingly, our MAE with a *single*-block decoder can perform strongly with fine-tuning (84.8%). Note that a single Transformer block is the minimal requirement to propagate information from visible tokens to mask tokens. Such a small decoder can further speed up training.

In Table 1b we study the decoder width (number of channels). We use 512-d by default, which performs well under fine-tuning and linear probing. A narrower decoder also works well with fine-tuning.

Overall, our default MAE decoder is lightweight. It has 8 blocks and a width of 512-d (gray in Table 1). It only has 9% FLOPs per token vs. ViT-L (24 blocks, 1024-d). As such, while the decoder processes all tokens, it is still a small fraction of the overall compute.

encoder	dec. depth	ft acc	hours	speedup
ViT-L, w/ [M]	8	84.2	42.4	-
ViT-L	8	84.9	15.4	2.8×
ViT-L	1	84.8	11.6	3.7×
ViT-H, w/ [M]	8	-	119.6 [†]	-
ViT-H	8	85.8	34.5	3.5×
ViT-H	1	85.9	29.3	4.1×

Table 2. **Wall-clock time** of our MAE training (800 epochs), benchmarked in 128 TPU-v3 cores with TensorFlow. The speedup is relative to the entry whose encoder has mask tokens (gray). The decoder width is 512, and the mask ratio is 75%. [†]: This entry is estimated by training ten epochs.

Mask token. An important design of our MAE is to skip the mask token [M] in the encoder and apply it later in the lightweight decoder. Table 1c studies this design.

If the encoder *uses* mask tokens, it performs *worse*: its accuracy drops by 14% in linear probing. In this case, there is a gap between pre-training and deploying: this encoder has a large portion of mask tokens in its input in pre-training, which does not exist in uncorrupted images. This gap may degrade accuracy in deployment. By removing the mask token from the encoder, we constrain the encoder to always see *real* patches and thus improve accuracy.

Moreover, by skipping the mask token in the encoder, we greatly reduce training computation. In Table 1c, we reduce the overall training FLOPs by 3.3×. This leads to a 2.8× wall-clock speedup in our implementation (see Table 2). The wall-clock speedup is even bigger (3.5–4.1×), for a smaller decoder (1-block), a larger encoder (ViT-H), or both. Note that the speedup can be >4× for a masking ratio of 75%, partially because the self-attention complexity is quadratic. In addition, memory is greatly reduced, which can enable training even larger models or speeding up more by large-batch training. The time and memory efficiency makes our MAE favorable for training very large models.

块数	ft	lin
1	84.8	65.5
2	84.9	70.0
4	84.9	71.9
8	84.9	73.5
12	84.4	73.3

(a) 解码器深度。较深解码器可提升线性探测准确率。

dim	ft	lin
128	84.9	69.1
256	84.8	71.3
512	84.9	73.5
768	84.4	73.1
1024	84.3	73.1

(b) 解码器宽度。解码器可比编码器（1024 维）更窄。

case	ft	lin	FLOPs
编码器含 [M]	84.2	59.6	3.3×
编码器不含 [M]	84.9	73.5	1×

(c) 掩码令牌。不使用掩码令牌的编码器更准确、更快（表 2）。

case	ft	lin
像素（未归一化）	84.9	73.5
像素（归一化）	85.4	73.9
PCA	84.6	72.3
dVAE 令牌	85.3	71.6

(d) 重建目标。以像素为重建目标有效。

case	ft	lin
none	84.0	65.7
裁剪，固定尺寸	84.7	73.1
裁剪，随机尺寸	84.9	73.5
裁剪 + 颜色扰动	84.3	71.9

(e) 数据增强。MAE 只需少量增强，甚至无需增强。

case	比例	ft	lin
随机	75	84.9	73.5
块状	50	83.9	72.3
块状	75	82.8	63.9
grid	75	84.0	66.0

(f) 掩码抽样。随机抽样最好，见图 6 可视化。

表 1. ImageNet-1K 上采用 ViT-L/16 的 MAE 消融实验。报告微调（ft）与线性探测（lin）准确率（%）。除特别说明外，默认设置为：解码器深度 8、宽度 512，重建目标为未归一化像素，数据增强为随机缩放裁剪，掩码比例 75%，预训练 800 轮。默认设置以灰色标注。

线性探测准确率随掩码比例持续增加，直至最佳点；准确率差距最长达 ~20%（54.6% 对 73.5%）。微调结果对比例较不敏感，40%–80% 的较大范围均表现良好。图 5 的所有微调结果都优于从零训练（82.5%）。

编码器	解码器深度	微调准确率	小时	加速比
ViT-L, w/ [M]	8	84.2	42.4	-
ViT-L	8	84.9	15.4	2.8×
ViT-L	1	84.8	11.6	3.7×
ViT-H, w/ [M]	8	-	119.6 [†]	-
ViT-H	8	85.8	34.5	3.5×
ViT-H	1	85.9	29.3	4.1×

表 2. MAE 训练（800 轮）的实际耗时，以 TensorFlow 在 128 个 TPU-v3 核上测量。加速相对于编码器使用掩码令牌的灰色条目。解码器宽度为 512，掩码比例为 75%。[†]: 此项由训练 10 轮估计。

掩码令牌。MAE 的重要设计是在编码器中跳过掩码令牌 [M]，稍后在轻量解码器中才使用。表 1c 研究这一设计。

若编码器使用掩码令牌，表现会变差：线性探测准确率下降 14%。此时预训练与应用之间存在差距：预训练时编码器输入中有大量掩码令牌，而完整图像中不存在这些令牌，可能降低应用时的准确率。移除编码器中的掩码令牌，可让编码器始终只看到真实图块，从而改善准确率。

此外，在编码器中跳过掩码令牌可大幅减少训练计算量。在表 1c 中，整体训练 FLOPs 降低了 3.3×。这在我们的实现中带来了 2.8× 的实际运行加速（见表 2）。使用更小的解码器（1 个块）、更大的编码器（ViT-H），或同时采用这两者时，实际加速更大（3.5–4.1×）。注意，75% 掩码比例下的加速可以达到 >4×，部分原因是自注意力的复杂度为二次方。此外，内存占用大幅降低，可以训练更大的模型，或通过大批量训练进一步提速。时间和内存效率使 MAE 特别适合训练超大型模型。

有趣的是，单块解码器的 MAE 仍能取得很强的微调结果（84.8%）。单个 Transformer 块是将信息从可见令牌传播到掩码令牌的最低要求。如此小的解码器可进一步加速训练。

表 1b 研究解码器宽度（通道数）。默认使用 512 维，微调和线性探测均表现良好。更窄的解码器微调时也有良好表现。

总的来说，默认 MAE 解码器很轻量，具有 8 个块、512 维宽度（表 1 灰色部分）。每令牌 FLOPs 仅为 ViT-L（24 个块、1024 维）的 9%。因此即使解码器处理全部令牌，它仍只占总计算量的一小部分。

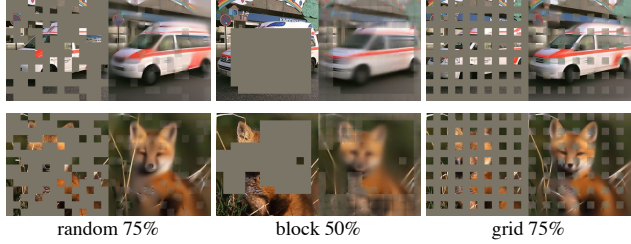


Figure 6. **Mask sampling strategies** determine the pretext task difficulty, influencing reconstruction quality and representations (Table 1f). Here each output is from an MAE trained with the specified masking strategy. Left: random sampling (our default). Middle: block-wise sampling [2] that removes large random blocks. Right: grid-wise sampling that keeps one of every four patches. Images are from the validation set.

Reconstruction target. We compare different reconstruction targets in Table 1d. Our results thus far are based on pixels without (per-patch) normalization. Using pixels *with* normalization improves accuracy. This per-patch normalization enhances the contrast locally. In another variant, we perform PCA in the patch space and use the largest PCA coefficients (96 here) as the target. Doing so degrades accuracy. Both experiments suggest that the high-frequency components are useful in our method.

We also compare an MAE variant that predicts *tokens*, the target used in BEiT [2]. Specifically for this variant, we use the DALLE pre-trained dVAE [50] as the tokenizer, following [2]. Here the MAE decoder predicts the token indices using cross-entropy loss. This tokenization improves fine-tuning accuracy by 0.4% *vs.* unnormalized pixels, but has no advantage *vs.* normalized pixels. It also reduces linear probing accuracy. In §5 we further show that tokenization is not necessary in transfer learning.

Our *pixel*-based MAE is much simpler than tokenization. The dVAE tokenizer requires one more pre-training stage, which may depend on extra data (250M images [50]). The dVAE encoder is a large convolutional network (40% FLOPs of ViT-L) and adds nontrivial overhead. Using pixels does not suffer from these problems.

Data augmentation. Table 1e studies the influence of data augmentation on our MAE pre-training.

Our MAE works well using *cropping-only* augmentation, either fixed-size or random-size (both having random horizontal flipping). Adding color jittering degrades the results and so we do not use it in other experiments.

Surprisingly, our MAE behaves decently even if using *no data augmentation* (only center-crop, no flipping). This property is dramatically different from contrastive learning and related methods [62, 23, 7, 21], which heavily rely on data augmentation. It was observed [21] that using cropping-only augmentation reduces the accuracy by 13%

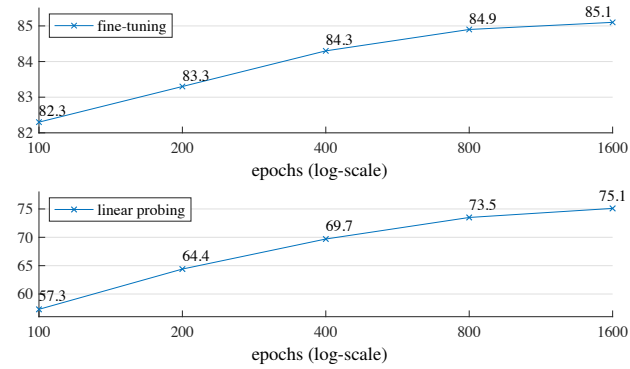


Figure 7. **Training schedules.** A longer training schedule gives a noticeable improvement. Here each point is a full training schedule. The model is ViT-L with the default setting in Table 1.

and 28% respectively for BYOL [21] and SimCLR [7]. In addition, there is no evidence that contrastive learning can work without augmentation: the two views of an image are the same and can easily satisfy a trivial solution.

In MAE, the role of data augmentation is mainly performed by random masking (ablated next). The masks are different for each iteration and so they generate new training samples regardless of data augmentation. The pretext task is made difficult by masking and requires less augmentation to regularize training.

Mask sampling strategy. In Table 1f we compare different mask sampling strategies, illustrated in Figure 6.

The *block-wise* masking strategy, proposed in [2], tends to remove large blocks (Figure 6 middle). Our MAE with block-wise masking works reasonably well at a ratio of 50%, but degrades at a ratio of 75%. This task is harder than that of random sampling, as a higher training loss is observed. The reconstruction is also blurrier.

We also study *grid-wise* sampling, which regularly keeps one of every four patches (Figure 6 right). This is an easier task and has lower training loss. The reconstruction is sharper. However, the representation quality is lower.

Simple random sampling works the best for our MAE. It allows for a higher masking ratio, which provides a greater speedup benefit while also enjoying good accuracy.

Training schedule. Our ablations thus far are based on 800-epoch pre-training. Figure 7 shows the influence of the training schedule length. The accuracy improves steadily with longer training. Indeed, we have not observed saturation of linear probing accuracy even at 1600 epochs. This behavior is unlike contrastive learning methods, *e.g.*, MoCo v3 [9] saturates at 300 epochs for ViT-L. Note that the MAE encoder only sees 25% of patches per epoch, while in contrastive learning the encoder sees 200% (two-crop) or even more (multi-crop) patches per epoch.

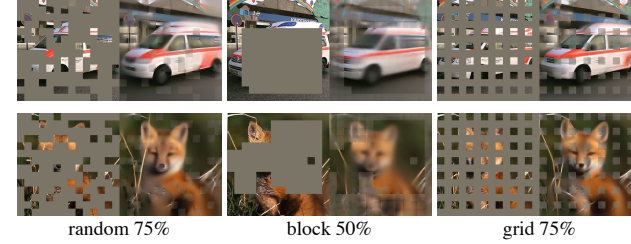


图 6. 掩码抽样策略决定前置任务难度，影响重建质量和表征（表 1f）。每个输出都来自按指定策略训练的 MAE。左：随机抽样（默认）；中：块状抽样 [2]，移除较大的随机块；右：网格抽样，每四个图块保留一个。图像来自验证集。

重建目标。表 1d 比较不同目标。此前结果使用不做逐图块归一化的像素。归一化像素可提升准确率，因为逐图块归一化增强了局部对比度。另一变体在图块空间中执行 PCA，并以最大的 PCA 系数（这里为 96 个）为目标，但准确率下降。这两个实验都说明高频成分对我们的方法有用。

我们还比较了预测令牌的 MAE 变体，即 BEiT [2] 采用的目标。具体而言，该变体遵循 [2]，以 DALLE 预训练的 dVAE [50] 作为分词器。MAE 解码器使用交叉熵损失预测令牌索引。与未归一化像素相比，这种令牌化将微调准确率提高 0.4%，但相对归一化像素没有优势，而且降低了线性探测准确率。第 5 节进一步说明，迁移学习无需令牌化。

像素式 MAE 比令牌化简单得多。dVAE 分词器还需要额外预训练阶段，可能依赖额外数据（2.5 亿图像 [50]）。dVAE 编码器是大型卷积网络（FLOPs 为 ViT-L 的 40%），带来明显开销。使用像素则没有这些问题。

数据增强。表 1e 研究增强对 MAE 预训练的影响。

MAE 仅采用裁剪增强时表现良好，无论固定尺寸还是随机尺寸（均有随机水平翻转）。加入颜色扰动会降低结果，因此其他实验不使用它。

令人意外的是，即使不使用数据增强（仅中心裁剪，不翻转），MAE 仍有尚好表现。这与高度依赖数据增强的对比学习及相关方法 [62, 23, 7, 21]，极不相同。已有研究 [21]发现，只有裁剪的增强会使准确率下降 13%

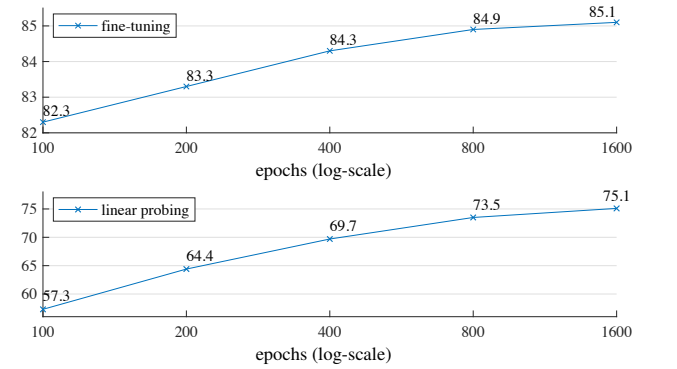


图 7. 训练日程。更长训练日程带来明显改进。每个点代表一次完整训练，模型为表 1 默认设置的 ViT-L。

与 28%，分别对应 BYOL [21] 和 SimCLR [7]。此外，没有证据说明对比学习无需增强也能工作：图像的两个视图相同，会轻易满足平凡解。

在 MAE 中，数据增强的作用主要由随机遮蔽承担（下文消融）。每次迭代的掩码不同，因此即使没有数据增强，也会产生新训练样本。遮蔽让前置任务变难，训练正则化因而需要更少增强。

掩码抽样策略。表 1f 比较不同策略，图 6 给出可视化。

[2]，提出的块状掩码策略倾向于移除大块区域（图 6 中）。MAE 在 50% 的块状掩码比例下表现尚好，但在 75% 时下降。该任务比随机抽样更难，训练损失更高，重建也更模糊。

我们还研究了每四个图块规律保留一个的网格抽样（图 6 右）。该任务更容易，训练损失更低，重建更清晰，但表征质量较差。

简单随机抽样最适合 MAE。它允许更高掩码比例，在保持良好准确率的同时获得更大的提速收益。

训练日程。此前消融采用 800 轮预训练。图 7 展示训练日程长度的影响：训练越久，准确率越稳步提高。事实上，即使到 1600 轮，线性探测准确率仍未饱和。这不同于对比学习方法，例如 MoCo v3 [9]的 ViT-L 在 300 轮时饱和。注意，MAE 编码器每轮仅看到 25% 的图块，而对比学习编码器每轮看到 200%（双裁剪）甚至更多（多裁剪）图块。

method	pre-train data	ViT-B	ViT-L	ViT-H	ViT-H ₄₄₈
scratch, our impl.	-	82.3	82.6	83.1	-
DINO [5]	IN1K	82.8	-	-	-
MoCo v3 [9]	IN1K	83.2	84.1	-	-
BEiT [2]	IN1K+DALLE	83.2	85.2	-	-
MAE	IN1K	<u>83.6</u>	<u>85.9</u>	<u>86.9</u>	87.8

Table 3. **Comparisons with previous results on ImageNet-1K.** The pre-training data is the ImageNet-1K training set (except the tokenizer in BEiT was pre-trained on 250M DALLE data [50]). All self-supervised methods are evaluated by end-to-end fine-tuning. The ViT models are B/16, L/16, H/14 [16]. The best for each column is underlined. All results are on an image size of 224, except for ViT-H with an extra result on 448. Here our MAE reconstructs normalized pixels and is pre-trained for 1600 epochs.

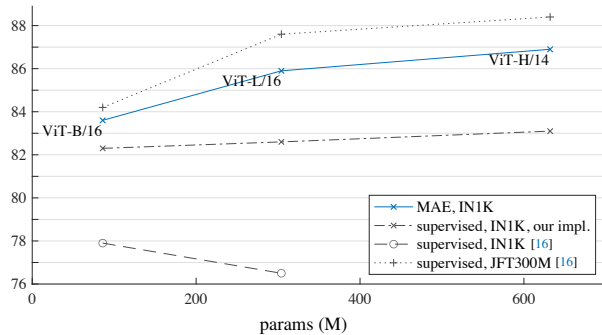


Figure 8. **MAE pre-training vs. supervised pre-training**, evaluated by fine-tuning in ImageNet-1K (224 size). We compare with the original ViT results [16] trained in IN1K or JFT300M.

4.2. Comparisons with Previous Results

Comparisons with self-supervised methods. In Table 3 we compare the fine-tuning results of self-supervised ViT models. For ViT-B, all methods perform closely. For ViT-L, the gaps among methods are bigger, suggesting that a challenge for bigger models is to reduce overfitting.

Our MAE can scale up easily and has shown steady improvement from bigger models. We obtain 86.9% accuracy using ViT-H (224 size). By fine-tuning with a 448 size, we achieve **87.8%** accuracy, *using only IN1K data*. The previous best accuracy, among all methods using only IN1K data, is 87.1% (512 size) [67], based on advanced networks. We improve over the state-of-the-art by a nontrivial margin in the highly competitive benchmark of IN1K (no external data). Our result is based on *vanilla* ViT, and we expect advanced networks will perform better.

Comparing with BEiT [2], our MAE is *more accurate* while being *simpler* and *faster*. Our method reconstructs pixels, in contrast to BEiT that predicts tokens: BEiT reported a 1.8% degradation [2] when reconstructing pixels with ViT-B.² We do not need dVAE pre-training. Moreover, our MAE is considerably faster ($3.5\times$ per epoch) than BEiT, for the reason as studied in Table 1c.

²We observed the degradation also in BEiT with ViT-L: it produces 85.2% (tokens) and 83.5% (pixels), reproduced from the official code.

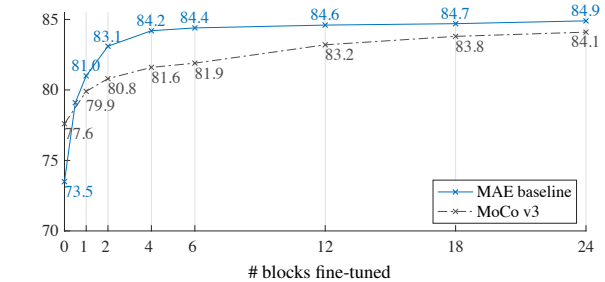


Figure 9. **Partial fine-tuning** results of ViT-L w.r.t. the number of fine-tuned Transformer blocks under the default settings from Table 1. Tuning 0 blocks is linear probing; 24 is full fine-tuning. Our MAE representations are less linearly separable, but are consistently better than MoCo v3 if one or more blocks are tuned.

The MAE models in Table 3 are pre-trained for 1600 epochs for better accuracy (Figure 7). Even so, our total pre-training time is *less* than the other methods when trained on the same hardware. For example, training ViT-L on 128 TPU-v3 cores, our MAE’s training time is 31 hours for 1600 epochs and MoCo v3’s is 36 hours for 300 epochs [9].

Comparisons with supervised pre-training. In the original ViT paper [16], ViT-L degrades when trained in IN1K. Our implementation of supervised training (see A.2) works better, but accuracy saturates. See Figure 8.

Our MAE pre-training, using only IN1K, can generalize better: the gain over training from scratch is bigger for higher-capacity models. It follows a trend similar to the JFT-300M *supervised* pre-training in [16]. This comparison shows that our MAE can help scale up model sizes.

4.3. Partial Fine-tuning

Table 1 shows that linear probing and fine-tuning results are largely *uncorrelated*. Linear probing has been a popular protocol in the past few years; however, it misses the opportunity of pursuing *strong but non-linear* features—which is indeed a strength of deep learning. As a middle ground, we study a *partial fine-tuning* protocol: fine-tune the last several layers while freezing the others. This protocol was also used in early works, *e.g.*, [65, 70, 42].

Figure 9 shows the results. Notably, fine-tuning only *one* Transformer block boosts the accuracy significantly from 73.5% to 81.0%. Moreover, if we fine-tune only “half” of the last block (*i.e.*, its MLP sub-block), we can get 79.1%, much better than linear probing. This variant is essentially fine-tuning an MLP head. Fine-tuning a few blocks (*e.g.*, 4 or 6) can achieve accuracy close to full fine-tuning.

In Figure 9 we also compare with MoCo v3 [9], a contrastive method with ViT-L results available. MoCo v3 has higher linear probing accuracy; however, all of its partial fine-tuning results are worse than MAE. The gap is 2.6% when tuning 4 blocks. While the MAE representations are less linearly separable, they are stronger *non-linear* features and perform well when a non-linear head is tuned.

方法	预训练数据	ViT-B	ViT-L	ViT-H	ViT-H ₄₄₈
从零训练, 我们的实现 -	-	82.3	82.6	83.1	-
DINO [5]	IN1K	82.8	-	-	-
MoCo v3 [9]	IN1K	83.2	84.1	-	-
BEiT [2]	IN1K+DALLE	83.2	85.2	-	-
MAE	IN1K	<u>83.6</u>	<u>85.9</u>	<u>86.9</u>	87.8

表 3. 与此前 ImageNet-1K 结果比较。预训练数据为 ImageNet-1K 训练集（但 BEiT 的分词器使用 2.5 亿张 DALLE 数据预训练[50]）。所有自监督方法都通过端到端微调评估。ViT 模型为 B/16、L/16、H/14 [16]。各列最佳结果加下划线。所有结果使用 224 图像尺寸，但 ViT-H 另提供 448 尺寸结果。这里的 MAE 重建归一化像素，预训练 1600 轮。

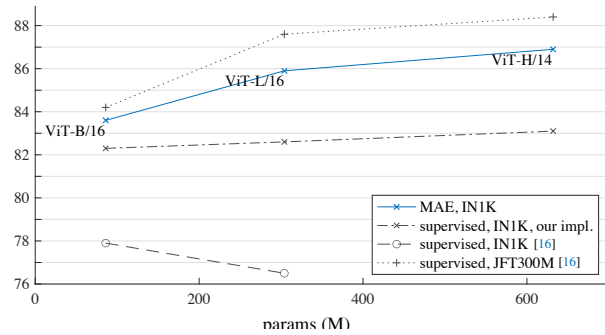


图 8. MAE 预训练与监督预训练的比较，通过 ImageNet-1K 上微调（224 尺寸）评估。比较原始 ViT [16] 在 IN1K 或 JFT300M 上训练的结果。

4.2. 与此前结果比较

与自监督方法比较。表 3 比较自监督 ViT 的微调结果。ViT-B 下，各方法相近；ViT-L 下差距变大，说明减少过拟合是大模型面临的一项挑战。

MAE 很容易扩大规模，并随着模型变大持续改进。使用 ViT-H（224 分辨率）获得 86.9% 准确率；以 448 分辨率微调后，在只使用 IN1K 数据的情况下达到 87.8%。此前所有仅使用 IN1K 数据的方法中，最高准确率为 87.1%（512 分辨率）[67]，基于更先进的网络。在竞争激烈的 IN1K 基准（无外部数据）上，我们以明显幅度超越此前最佳结果。我们的结果基于普通 ViT，预计更先进的网络还会更好。

与 BEiT [2], 相比, MAE 更准确、更简单且更快。我们的方法重建像素, 而 BEiT 预测令牌: BEiT 报告 ViT-B 重建像素时性能下降 1.8% [2]。² 我们无需 dVAE 预训练。此外, 基于表 1c 分析的原因, MAE 每轮训练显著快于 BEiT ($3.5\times$)。

²我们在 ViT-L 的 BEiT 中也观察到下降: 按官方代码复现, 令牌目标为 85.2%, 像素目标为 83.5%。

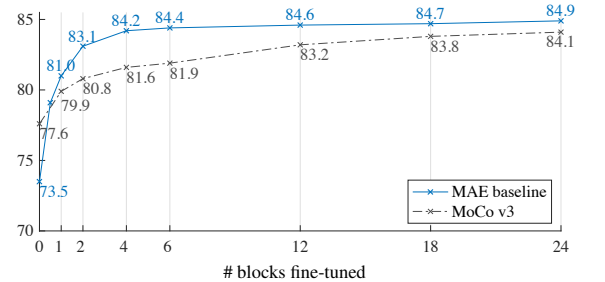


图 9. 在表 1 默认设置下，ViT-L 的部分微调结果随所微调 Transformer 块数的变化。0 个块即线性探测；24 个块即完整微调。MAE 表征的线性可分性较弱，但只要微调至少一个块，就始终优于 MoCo v3。

为了更高准确率，表 3 中的 MAE 预训练 1600 轮（图 7）。即便如此，在相同硬件上，总预训练时间仍少于其他方法。例如在 128 个 TPU-v3 核上训练 ViT-L，MAE 的 1600 轮耗时 31 小时，而 MoCo v3 的 300 轮耗时 36 小时 [9]。与监督预训练比较。原始 ViT 论文 [16], 中, ViT-L 在 IN1K 上训练时性能下降。我们的监督训练实现（见 A.2）更好，但准确率趋于饱和，见图 8。

MAE 仅用 IN1K 预训练，就能更好泛化：相对从零训练，容量更大的模型获得更大收益。趋势类似于 [16]中 JFT-300M 监督预训练，说明 MAE 有助于扩大模型规模。

4.3. 部分微调

表 1 表明，线性探测与微调结果在很大程度上不相关。线性探测在过去几年中很流行，但它错失了探索强大非线性特征的机会，而非线性正是深度学习的优势。作为折中，我们研究部分微调协议：冻结其余层，只微调最后几层。早期工作（如 [65, 70, 42]）也使用过这种协议。

图 9 给出结果。值得注意的是，仅微调一个 Transformer 块便将准确率从 73.5% 大幅提升至 81.0%。即使只微调最后一个块的“一半”（MLP 子模块），也能达到 79.1%，显著优于线性探测。该变体本质上是在微调一个 MLP 头。微调少数几个块（如 4 或 6 个），即可接近完整微调的准确率。

图 9 还与提供 ViT-L 结果的对比方法 MoCo v 3 [9], 比较。MoCo v3 的线性探测准确率更高，但所有部分微调结果均逊于 MAE。微调 4 个块时差距为 2.6%。虽然 MAE 表征的线性可分性较弱，但它具有更强的非线性特征，微调非线性头后性能良好。

method	pre-train data	AP ^{box}		AP ^{mask}	
		ViT-B	ViT-L	ViT-B	ViT-L
supervised	IN1K w/ labels	47.9	49.3	42.9	43.9
MoCo v3	IN1K	47.9	49.3	42.7	44.0
BEiT	IN1K+DALLE	49.8	53.3	44.4	47.1
MAE	IN1K	50.3	53.3	44.9	47.2

Table 4. **COCO object detection and segmentation** using a ViT Mask R-CNN baseline. All entries are based on our implementation. Self-supervised entries use IN1K data *without* labels. Mask AP follows a similar trend as box AP.

These observations suggest that linear separability is not the sole metric for evaluating representation quality. It has also been observed (*e.g.*, [8]) that linear probing is not well correlated with transfer learning performance, *e.g.*, for object detection. To our knowledge, linear evaluation is not often used in NLP for benchmarking pre-training.

5. Transfer Learning Experiments

We evaluate transfer learning in downstream tasks using the pre-trained models in Table 3.

Object detection and segmentation. We fine-tune Mask R-CNN [24] end-to-end on COCO [37]. The ViT backbone is adapted for use with FPN [36] (see A.3). We apply this approach for all entries in Table 4. We report box AP for object detection and mask AP for instance segmentation.

Compared to supervised pre-training, our MAE performs better under all configurations (Table 4). With the smaller ViT-B, our MAE is 2.4 points higher than *supervised* pre-training (50.3 *vs.* 47.9, AP^{box}). More significantly, with the larger ViT-L, our MAE pre-training outperforms supervised pre-training by 4.0 points (53.3 *vs.* 49.3).

The *pixel*-based MAE is better than or on par with the *token*-based BEiT, while MAE is much simpler and faster. Both MAE and BEiT are better than MoCo v3 and MoCo v3 is on par with supervised pre-training.

Semantic segmentation. We experiment on ADE20K [72] using UperNet [63] (see A.4). Table 5 shows that our pre-training significantly improves results over *supervised* pre-training, *e.g.*, by 3.7 points for ViT-L. Our pixel-based MAE also outperforms the token-based BEiT. These observations are consistent with those in COCO.

Classification tasks. Table 6 studies transfer learning on the iNaturalists [56] and Places [71] tasks (see A.5). On iNat, our method shows strong scaling behavior: accuracy improves considerably with bigger models. Our results surpass the previous best results *by large margins*. On Places, our MAE outperforms the previous best results [19, 40], which were obtained via pre-training on billions of images.

Pixels vs. tokens. Table 7 compares pixels *vs.* tokens as the MAE reconstruction target. While using dVAE tokens is better than using *unnormalized* pixels, it is statistically similar to using *normalized* pixels across all cases we tested. It again shows that tokenization is not necessary for our MAE.

method	pre-train data	ViT-B	ViT-L
supervised	IN1K w/ labels	47.4	49.9
MoCo v3	IN1K	47.3	49.1
BEiT	IN1K+DALLE	47.1	53.3
MAE	IN1K	48.1	53.6

Table 5. **ADE20K semantic segmentation** (mIoU) using UperNet. BEiT results are reproduced using the official code. Other entries are based on our implementation. Self-supervised entries use IN1K data *without* labels.

dataset	ViT-B	ViT-L	ViT-H	ViT-H ₄₄₈	prev best
iNat 2017	70.5	75.7	79.3	83.4	75.4 [55]
iNat 2018	75.4	80.1	83.0	86.8	81.2 [54]
iNat 2019	80.5	83.4	85.7	88.3	84.1 [54]
Places205	63.9	65.8	65.9	66.8	66.0 [19] [†]
Places365	57.9	59.4	59.8	60.3	58.0 [40] [‡]

Table 6. **Transfer learning accuracy on classification datasets**, using MAE pre-trained on IN1K and then fine-tuned. We provide system-level comparisons with the previous best results.

[†]: pre-trained on 1 billion images. [‡]: pre-trained on 3.5 billion images.

	IN1K			COCO		ADE20K	
	ViT-B	ViT-L	ViT-H	ViT-B	ViT-L	ViT-B	ViT-L
pixel (w/o norm)	83.3	85.1	86.2	49.5	52.8	48.0	51.8
pixel (w/ norm)	83.6	85.9	86.9	50.3	53.3	48.1	53.6
dVAE token	83.6	85.7	86.9	50.3	53.2	48.1	53.4
△	0.0	-0.2	0.0	0.0	-0.1	0.0	-0.2

Table 7. **Pixels vs. tokens** as the MAE reconstruction target. △ is the difference between using dVAE tokens and using normalized pixels. The difference is statistically insignificant.

6. Discussion and Conclusion

Simple algorithms that scale well are the core of deep learning. In NLP, simple self-supervised learning methods (*e.g.*, [47, 14, 48, 4]) enable benefits from exponentially scaling models. In computer vision, practical pre-training paradigms are dominantly supervised (*e.g.* [33, 51, 25, 16]) despite progress in self-supervised learning. In this study, we observe on ImageNet and in transfer learning that an autoencoder—a simple self-supervised method similar to techniques in NLP—provides scalable benefits. Self-supervised learning in vision may now be embarking on a similar trajectory as in NLP.

On the other hand, we note that images and languages are *signals of a different nature* and this difference must be addressed carefully. Images are merely recorded light *without* a semantic decomposition into the visual analogue of words. Instead of attempting to remove objects, we remove random patches that most likely do *not* form a semantic segment. Likewise, our MAE reconstructs pixels, which are *not* semantic entities. Nevertheless, we observe (*e.g.*, Figure 4) that our MAE infers complex, holistic reconstructions, suggesting it has learned numerous visual concepts, *i.e.*, semantics. We hypothesize that this behavior occurs by way of a rich hidden representation inside the MAE. We hope this perspective will inspire future work.

方法	预训练数据	AP ^{box}		AP ^{mask}	
		ViT-B	ViT-L	ViT-B	ViT-L
监督	带标签 IN1K	47.9	49.3	42.9	43.9
MoCo v3	IN1K	47.9	49.3	42.7	44.0
BEiT	IN1K+DALLE	49.8	53.3	44.4	47.1
MAE	IN1K	50.3	53.3	44.9	47.2

表 4. 使用 ViT Mask R-CNN 基线的 COCO 目标检测与分割。所有条目基于我们的实现，自监督条目使用无标签 IN1K 数据。掩码 AP 与框 AP 趋势相似。

这些观察说明，线性可分性并非评估表征质量的唯一指标。已有研究（如 [8]）也发现，线性探测与迁移学习性能（如目标检测）相关性不强。据我们所知，NLP 也不常用线性评估来衡量预训练。

5. 迁移学习实验

使用表 3 预训练模型评估下游迁移学习。

目标检测与分割。我们在 COCO [37]上端到端微调 Mask R-CNN [24]，将 ViT 骨干改造为可配合 FPN [36]使用（见 A.3），表 4 的所有条目均采用此方法。报告目标检测的框 AP 和实例分割的掩码 AP。

相对监督预训练，MAE 在全部配置中更好（表 4）。使用较小的 ViT-B 时，比监督预训练高 2.4 个点（50.3 对 47.9, AP^{box}）。更显著的是，较大的 ViT-L 提高了 4.0 个点（53.3 对 49.3）。

像素式 MAE 优于或持平于令牌式 BEiT，同时更简单、更快。MAE 与 BEiT 均优于 MoCo v3，后者与监督预训练相近。

语义分割。我们在 ADE20K [72]上使用 UperNet [63] 实验（见 A.4）。表 5 显示，相对监督预训练，我们显著改善结果，例如 ViT-L 提高 3.7 个点。像素式 MAE 也超过令牌式 BEiT，与 COCO 中的观察一致。

分类任务。表 6 研究在 iNaturalist [56] 与 Places [71] 任务上的迁移（见 A.5）。在 iNat 上，模型越大，准确率越显著提高，体现良好扩展性，并大幅超过此前最佳结果。在 Places 上，MAE 超过了使用数十亿图像预训练得到的此前最佳结果 [19, 40]。

像素与令牌。表 7 比较二者作为 MAE 重建目标。dVAE 令牌虽然优于未归一化像素，但在全部测试中与归一化像素的差异在统计上不显著。这再次说明，MAE 无需令牌化。

方法	预训练数据	ViT-B	ViT-L
监督	带标签 IN1K	47.4	49.9
MoCo v3	IN1K	47.3	49.1
BEiT	IN1K+DALLE	47.1	53.3
MAE	IN1K	48.1	53.6

表 5. 使用 UperNet 的 ADE20K 语义分割（mIoU）。BEiT 用官方代码复现，其余条目基于我们的实现。自监督条目使用无标签 IN1K 数据。

数据集	ViT-B	ViT-L	ViT-H	ViT-H ₄₄₈	此前最佳
iNat 2017	70.5	75.7	79.3	83.4	75.4 [55]
iNat 2018	75.4	80.1	83.0	86.8	81.2 [54]
iNat 2019	80.5	83.4	85.7	88.3	84.1 [54]
Places205	63.9	65.8	65.9	66.8	66.0 [19] [†]
Places365	57.9	59.4	59.8	60.3	58.0 [40] [‡]

表 6. 分类数据集上的迁移准确率：先在 IN1K 预训练 MAE，再微调。与此前最佳结果进行系统级比较。

[†]: 用 10 亿图像预训练。 [‡]: 用 35 亿图像预训练。

	IN1K			COCO		ADE20K	
	ViT-B	ViT-L	ViT-H	ViT-B	ViT-L	ViT-B	ViT-L
像素（未归一化）	83.3	85.1	86.2	49.5	52.8	48.0	51.8
像素（归一化）	83.6	85.9	86.9	50.3	53.3	48.1	53.6
dVAE 令牌	83.6	85.7	86.9	50.3	53.2	48.1	53.4
△	0.0	-0.2	0.0	0.0	-0.1	0.0	-0.2

表 7. 像素与令牌作为 MAE 重建目标的比较。△ 为 dVAE 令牌与归一化像素的结果差，统计上不显著。

6. 讨论与结论

能够良好扩展的简单算法是深度学习的核心。在 NLP 中，简单的自监督学习方法（如 [47, 14, 48, 4]）能从模型规模的指数增长中获益。在计算机视觉中，尽管自监督学习取得进展，实际预训练范式仍以监督方法为主（如 [33, 51, 25, 16]）。本研究在 ImageNet 与迁移学习中观察到，自编码器这一与 NLP 技术相似的简单自监督方法，可提供随规模扩展的收益。视觉自监督学习可能正开始踏上与 NLP 类似的道路。

另一方面，图像与语言是性质不同的信号，必须仔细处理这种差异。图像只是被记录的光，并未像词语那样被语义分解为相应的视觉单元。我们并不尝试移除物体，而是移除很可能并不构成语义片段的随机图块。同样，MAE 重建的是像素，而非语义实体。不过，我们观察到（如图 4）MAE 能推断复杂且整体协调的重建，说明它学到了许多视觉概念，即语义。我们推测，这种行为来自 MAE 内部丰富的隐藏表征。希望这一视角能启发未来研究。

Broader impacts. The proposed method predicts content based on learned statistics of the training dataset and as such will reflect biases in those data, including ones with negative societal impacts. The model may generate inexistent content. These issues warrant further research and consideration when building upon this work to generate images.

References

- [1] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv:1607.06450*, 2016.
- [2] Hangbo Bao, Li Dong, and Furu Wei. BEiT: BERT pre-training of image transformers. *arXiv:2106.08254*, 2021. *Accessed in June 2021*.
- [3] Suzanna Becker and Geoffrey E Hinton. Self-organizing neural network that discovers surfaces in random-dot stereograms. *Nature*, 1992.
- [4] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In *NeurIPS*, 2020.
- [5] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *ICCV*, 2021.
- [6] Mark Chen, Alec Radford, Rewon Child, Jeffrey Wu, Heewoo Jun, David Luan, and Ilya Sutskever. Generative pretraining from pixels. In *ICML*, 2020.
- [7] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *ICML*, 2020.
- [8] Xinlei Chen and Kaiming He. Exploring simple Siamese representation learning. In *CVPR*, 2021.
- [9] Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised Vision Transformers. In *ICCV*, 2021.
- [10] Kevin Clark, Minh-Thang Luong, Quoc V Le, and Christopher D Manning. ELECTRA: Pre-training text encoders as discriminators rather than generators. In *ICLR*, 2020.
- [11] Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine learning*, 1995.
- [12] Ekin D Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V Le. Randaugment: Practical automated data augmentation with a reduced search space. In *CVPR Workshops*, 2020.
- [13] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *CVPR*, 2009.
- [14] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *NAACL*, 2019.
- [15] Carl Doersch, Abhinav Gupta, and Alexei A Efros. Unsupervised visual representation learning by context prediction. In *ICCV*, 2015.
- [16] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob

Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.

- [17] Spyros Gidaris, Praveer Singh, and Nikos Komodakis. Unsupervised representation learning by predicting image rotations. In *ICLR*, 2018.
- [18] Xavier Glorot and Yoshua Bengio. Understanding the difficulty of training deep feedforward neural networks. In *AISTATS*, 2010.
- [19] Priya Goyal, Mathilde Caron, Benjamin Lefaudeaux, Min Xu, Pengchao Wang, Vivek Pai, Mannat Singh, Vitaliy Liptchinsky, Ishan Misra, Armand Joulin, and Piotr Bojanowski. Self-supervised pretraining of visual features in the wild. *arXiv:2103.01988*, 2021.
- [20] Priya Goyal, Piotr Dollár, Ross Girshick, Pieter Noordhuis, Lukasz Wesolowski, Aapo Kyröla, Andrew Tulloch, Yangqing Jia, and Kaiming He. Accurate, large minibatch SGD: Training ImageNet in 1 hour. *arXiv:1706.02677*, 2017.
- [21] Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Remi Munos, and Michal Valko. Bootstrap your own latent - a new approach to self-supervised learning. In *NeurIPS*, 2020.
- [22] Raia Hadsell, Sumit Chopra, and Yann LeCun. Dimensionality reduction by learning an invariant mapping. In *CVPR*, 2006.
- [23] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *CVPR*, 2020.
- [24] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask R-CNN. In *ICCV*, 2017.
- [25] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2016.
- [26] Dan Hendrycks, Steven Basart, Norman Mu, Saurav Kadavath, Frank Wang, Evan Dorundo, Rahul Desai, Tyler Zhu, Samyak Parajuli, Mike Guo, et al. The many faces of robustness: A critical analysis of out-of-distribution generalization. In *ICCV*, 2021.
- [27] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *ICLR*, 2019.
- [28] Dan Hendrycks, Kevin Zhao, Steven Basart, Jacob Steinhardt, and Dawn Song. Natural adversarial examples. In *CVPR*, 2021.
- [29] Geoffrey E Hinton and Richard S Zemel. Autoencoders, minimum description length, and helmholtz free energy. In *NeurIPS*, 1994.
- [30] Gao Huang, Yu Sun, Zhuang Liu, Daniel Sedra, and Kilian Q Weinberger. Deep networks with stochastic depth. In *ECCV*, 2016.
- [31] Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *ICML*, 2015.
- [32] Insoo Kim, Seungju Han, Ji-won Baek, Seong-Jin Park, Jae-Joon Han, and Jinwoo Shin. Quality-agnostic image recognition via invertible decoder. In *CVPR*, 2021.
- [33] Alex Krizhevsky, Ilya Sutskever, and Geoff Hinton. Imagenet classification with deep convolutional neural networks. In *NeurIPS*, 2012.
- [34] Yann LeCun, Bernhard Boser, John S Denker, Donnie Hender-son, Richard E Howard, Wayne Hubbard, and Lawrence D Jackel. Backpropagation applied to handwritten zip code recognition. *Neural computation*, 1989.
- [35] Yanghao Li, Saining Xie, Xinlei Chen, Piotr Dollár, Kaiming He, and Ross Girshick. Benchmarking detection transfer learning with vision transformers. In *preparation*, 2021.

更广泛影响。该方法根据训练数据集学到的统计规律预测内容，因此会反映数据偏见，包括具有负面社会影响的偏见。模型可能生成不存在的内容。基于本工作进行图像生成时，这些问题值得进一步研究与考虑。

参考文献

- [1] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. arXiv:1607.06450, 2016.[2] Hangbo Bao, Li Dong, and Furu Wei. BEiT: BERT pre-training of image transformers. arXiv:2106.08254, 2021. Accessed in June 2021.[3] Suzanna Becker and Geoffrey E Hinton. Self-organizing neural network that discovers surfaces in random-dot stereograms. Nature, 1992.[4] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In NeurIPS, 2020.[5] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In ICCV, 2021.[6] Mark Chen, Alec Radford, Rewon Child, Jeffrey Wu, Heewoo Jun, David Luan, and Ilya Sutskever. Generative pretraining from pixels. In ICML, 2020.[7] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In ICML, 2020.[8] Xinlei Chen and Kaiming He. Exploring simple Siamese representation learning. In CVPR, 2021.[9] Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised Vision Transformers. In ICCV, 2021.[10] Kevin Clark, Minh-Thang Luong, Quoc V Le, and Christopher D Manning. ELECTRA: Pre-training text encoders as discriminators rather than generators. In ICLR, 2020.[11] Corinna Cortes and Vladimir Vapnik. Support-vector networks. Machine learning, 1995.[12] Ekin D Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V Le. Randaugment: Practical automated data augmentation with a reduced search space. In CVPR Workshops, 2020.[13] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In CVPR, 2009.[14] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In NAACL, 2019.[15] Carl Doersch, Abhinav Gupta, and Alexei A Efros. Unsupervised visual representation learning by context prediction. In ICCV, 2015.[16] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob

Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In ICLR, 2021.[17] Spyros Gidaris, Praveer Singh, and Nikos Komodakis. Unsuper-vised representation learning by predicting image rotations. In ICLR, 2018.[18] Xavier Glorot and Yoshua Bengio. Understanding the difficulty of training deep feedforward neural networks. In AISTATS, 2010.[19] Priya Goyal, Mathilde Caron, Benjamin Lefaudeaux, Min Xu, Pengchao Wang, Vivek Pai, Mannat Singh, Vitaliy Liptchinsky, Ishan Misra, Armand Joulin, and Piotr Bojanowski. Self-supervised pretraining of visual features in the wild. arXiv:2103.01988, 2021.[20] Priya Goyal, Piotr Dollár, Ross Girshick, Pieter Noordhuis, Lukasz Wesolowski, Aapo Kyröla, Andrew Tulloch, Yangqing Jia, and Kaiming He. Accurate, large minibatch SGD: Training ImageNet in 1 hour. arXiv:1706.02677, 2017.[21] Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Remi Munos, and Michal Valko. Bootstrap your own latent - a new approach to self-supervised learning. In NeurIPS, 2020.[22] Raia Hadsell, Sumit Chopra, and Yann LeCun. Dimensionality reduction by learning an invariant mapping. In CVPR, 2006.[23] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In CVPR, 2020.[24] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask R-CNN. In ICCV, 2017.[25] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In CVPR, 2016.[26] Dan Hendrycks, Steven Basart, Norman Mu, Saurav Kadavath, Frank Wang, Evan Dorundo, Rahul Desai, Tyler Zhu, Samyak Parajuli, Mike Guo, et al. The many faces of robustness: A critical analysis of out-of-distribution generalization. In ICCV, 2021.[27] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In ICLR, 2019.[28] Dan Hendrycks, Kevin Zhao, Steven Basart, Jacob Steinhardt, and Dawn Song. Natural adversarial examples. In CVPR, 2021.[29] Geoffrey E Hinton and Richard S Zemel. Autoencoders, minimum description length, and helmholtz free energy. In NeurIPS, 1994.[30] Gao Huang, Yu Sun, Zhuang Liu, Daniel Sedra, and Kilian Q Weinberger. Deep networks with stochastic depth. In ECCV, 2016.[31] Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In ICML, 2015.[32] Insoo Kim, Seungju Han, Ji-won Baek, Seong-Jin Park, Jae-Joon Han, and Jinwoo Shin. Quality-agnostic image recognition via invertible decoder. In CVPR, 2021.[33] Alex Krizhevsky, Ilya Sutskever, and Geoff Hinton. Imagenet classification with deep convolutional neural networks. In NeurIPS, 2012.[34] Yann LeCun, Bernhard Boser, John S Denker, Donnie Hender-son, Richard E Howard, Wayne Hubbard, and Lawrence D Jackel. Backpropagation applied to handwritten zip code recognition. Neural computation, 1989.[35] Yanghao Li, Saining Xie, Xinlei Chen, Piotr Dollár, Kaiming He, and Ross Girshick. Benchmarking detection transfer learning with vision transformers. In preparation, 2021.

- [36] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. Feature pyramid networks for object detection. In *CVPR*, 2017.
- [37] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft COCO: Common objects in context. In *ECCV*, 2014.
- [38] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *ICLR*, 2017.
- [39] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *ICLR*, 2019.
- [40] Dhruv Mahajan, Ross Girshick, Vignesh Ramanathan, Kaiming He, Manohar Paluri, Yixuan Li, Ashwin Bharambe, and Laurens van der Maaten. Exploring the limits of weakly supervised pre-training. In *ECCV*, 2018.
- [41] Xiaofeng Mao, Gege Qi, Yuefeng Chen, Xiaodan Li, Ranjie Duan, Shaokai Ye, Yuan He, and Hui Xue. Towards robust vision transformer. *arXiv:2105.07926*, 2021.
- [42] Mehdi Noroozi and Paolo Favaro. Unsupervised learning of visual representations by solving jigsaw puzzles. In *ECCV*, 2016.
- [43] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv:1807.03748*, 2018.
- [44] Aaron van den Oord, Oriol Vinyals, and Koray Kavukcuoglu. Neural discrete representation learning. In *NeurIPS*, 2017.
- [45] Deepak Pathak, Ross Girshick, Piotr Dollár, Trevor Darrell, and Bharath Hariharan. Learning features by watching objects move. In *CVPR*, 2017.
- [46] Deepak Pathak, Philipp Krahenbuhl, Jeff Donahue, Trevor Darrell, and Alexei A Efros. Context encoders: Feature learning by inpainting. In *CVPR*, 2016.
- [47] Alec Radford, Karthik Narasimhan, Tim Salimans, and Ilya Sutskever. Improving language understanding by generative pre-training. 2018.
- [48] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. 2019.
- [49] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *JMLR*, 2020.
- [50] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *ICML*, 2021.
- [51] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. In *ICLR*, 2015.
- [52] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jonathon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *CVPR*, 2016.
- [53] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *ICML*, 2021.
- [54] Hugo Touvron, Alexandre Sablayrolles, Matthijs Douze, Matthieu Cord, and Hervé Jégou. Graft: Learning fine-grained image representations with coarse labels. In *ICCV*, 2021.
- [55] Hugo Touvron, Andrea Vedaldi, Matthijs Douze, and Hervé Jégou. Fixing the train-test resolution discrepancy. *arXiv:1906.06423*, 2019.
- [56] Grant Van Horn, Oisin Mac Aodha, Yang Song, Yin Cui, Chen Sun, Alex Shepard, Hartwig Adam, Pietro Perona, and Serge Belongie. The iNaturalist species classification and detection dataset. In *CVPR*, 2018.
- [57] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *NeurIPS*, 2017.
- [58] Pascal Vincent, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol. Extracting and composing robust features with denoising autoencoders. In *ICML*, 2008.
- [59] Pascal Vincent, Hugo Larochelle, Isabelle Lajoie, Yoshua Bengio, Pierre-Antoine Manzagol, and Léon Bottou. Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion. *JMLR*, 2010.
- [60] Haohan Wang, Songwei Ge, Zachary Lipton, and Eric P Xing. Learning robust global representations by penalizing local predictive power. In *NeurIPS*, 2019.
- [61] Xiaolong Wang and Abhinav Gupta. Unsupervised learning of visual representations using videos. In *ICCV*, 2015.
- [62] Zhirong Wu, Yuanjun Xiong, Stella Yu, and Dahua Lin. Unsupervised feature learning via non-parametric instance discrimination. In *CVPR*, 2018.
- [63] Tete Xiao, Yingcheng Liu, Bolei Zhou, Yuning Jiang, and Jian Sun. Unified perceptual parsing for scene understanding. In *ECCV*, 2018.
- [64] Tete Xiao, Mannat Singh, Eric Mintun, Trevor Darrell, Piotr Dollár, and Ross Girshick. Early convolutions help transformers see better. In *NeurIPS*, 2021.
- [65] Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson. How transferable are features in deep neural networks? In *NeurIPS*, 2014.
- [66] Yang You, Igor Gitman, and Boris Ginsburg. Large batch training of convolutional networks. *arXiv:1708.03888*, 2017.
- [67] Li Yuan, Qibin Hou, Zihang Jiang, Jiashi Feng, and Shuicheng Yan. VOLO: Vision outlooker for visual recognition. *arXiv:2106.13112*, 2021.
- [68] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. Cutmix: Regularization strategy to train strong classifiers with localizable features. In *ICCV*, 2019.
- [69] Hongyi Zhang, Moustapha Cisse, Yann N Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. In *ICLR*, 2018.
- [70] Richard Zhang, Phillip Isola, and Alexei A Efros. Colorful image colorization. In *ECCV*, 2016.
- [71] Bolei Zhou, Agata Lapedriza, Jianxiong Xiao, Antonio Torralba, and Aude Oliva. Learning deep features for scene recognition using Places database. In *NeurIPS*, 2014.
- [72] Bolei Zhou, Hang Zhao, Xavier Puig, Tete Xiao, Sanja Fidler, Adela Barriuso, and Antonio Torralba. Semantic understanding of scenes through the ADE20K dataset. *IJCV*, 2019.

- [36] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. Feature pyramid networks for object detection. In *CVPR*, 2017.
- [37] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft COCO: Common objects in context. In *ECCV*, 2014.
- [38] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *ICLR*, 2017.
- [39] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *ICLR*, 2019.
- [40] Dhruv Mahajan, Ross Girshick, Vignesh Ramanathan, Kaiming He, Manohar Paluri, Yixuan Li, Ashwin Bharambe, and Laurens van der Maaten. Exploring the limits of weakly supervised pre-training. In *ECCV*, 2018.
- [41] Xiaofeng Mao, Gege Qi, Yuefeng Chen, Xiaodan Li, Ranjie Duan, Shaokai Ye, Yuan He, and Hui Xue. Towards robust vision transformer. *arXiv:2105.07926*, 2021.
- [42] Mehdi Noroozi and Paolo Favaro. Unsupervised learning of visual representations by solving jigsaw puzzles. In *ECCV*, 2016.
- [43] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv:1807.03748*, 2018.

- [44] Aaron van den Oord, Oriol Vinyals, and Koray Kavukcuoglu. Neural discrete representation learning. In *NeurIPS*, 2017.
- [45] Deepak Pathak, Ross Girshick, Piotr Dollár, Trevor Darrell, and Bharath Hariharan. Learning features by watching objects move. In *CVPR*, 2017.
- [46] Deepak Pathak, Philipp Krahenbuhl, Jeff Donahue, Trevor Darrell, and Alexei A Efros. Context encoders: Feature learning by inpainting. In *CVPR*, 2016.
- [47] Alec Radford, Karthik Narasimhan, Tim Salimans, and Ilya Sutskever. Improving language understanding by generative pre-training. 2018.
- [48] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. 2019.
- [49] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *JMLR*, 2020.
- [50] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *ICML*, 2021.
- [51] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. In *ICLR*, 2015.
- [52] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jonathon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *CVPR*, 2016.
- [53] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *ICML*, 2021.
- [54] Hugo Touvron, Alexandre Sablayrolles, Matthijs Douze, Matthieu Cord, and Hervé Jégou. Graft: Learning fine-grained image representations with coarse labels. In *ICCV*, 2021.
- [55] Hugo Touvron, Andrea Vedaldi, Matthijs Douze, and Hervé Jégou. Fixing the train-test resolution discrepancy. *arXiv:1906.06423*, 2019.

- [56] Grant Van Horn, Oisin Mac Aodha, Yang Song, Yin Cui, Chen Sun, Alex Shepard, Hartwig Adam, Pietro Perona, and Serge Belongie. The iNaturalist species classification and detection dataset. In *CVPR*, 2018.
- [57] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *NeurIPS*, 2017.
- [58] Pascal Vincent, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol. Extracting and composing robust features with denoising autoencoders. In *ICML*, 2008.
- [59] Pascal Vincent, Hugo Larochelle, Isabelle Lajoie, Yoshua Bengio, Pierre-Antoine Manzagol, and Léon Bottou. Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion. *JMLR*, 2010.
- [60] Haohan Wang, Songwei Ge, Zachary Lipton, and Eric P Xing. Learning robust global representations by penalizing local predictive power. In *NeurIPS*, 2019.
- [61] Xiaolong Wang and Abhinav Gupta. Unsupervised learning of visual representations using videos. In *ICCV*, 2015.
- [62] Zhirong Wu, Yuanjun Xiong, Stella Yu, and Dahua Lin. Unsupervised feature learning via non-parametric instance discrimination. In *CVPR*, 2018.
- [63] Tete Xiao, Yingcheng Liu, Bolei Zhou, Yuning Jiang, and Jian Sun. Unified perceptual parsing for scene understanding. In *ECCV*, 2018.
- [64] Tete Xiao, Mannat Singh, Eric Mintun, Trevor Darrell, Piotr Dollár, and Ross Girshick. Early convolutions help transformers see better. In *NeurIPS*, 2021.
- [65] Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson. How transferable are features in deep neural networks? In *NeurIPS*, 2014.
- [66] Yang You, Igor Gitman, and Boris Ginsburg. Large batch training of convolutional networks. *arXiv:1708.03888*, 2017.
- [67] Li Yuan, Qibin Hou, Zihang Jiang, Jiashi Feng, and Shuicheng Yan. VOLO: Vision outlooker for visual recognition. *arXiv:2106.13112*, 2021.
- [68] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. Cutmix: Regularization strategy to train strong classifiers with localizable features. In *ICCV*, 2019.
- [69] Hongyi Zhang, Moustapha Cisse, Yann N Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. In *ICLR*, 2018.
- [70] Richard Zhang, Phillip Isola, and Alexei A Efros. Colorful image colorization. In *ECCV*, 2016.
- [71] Bolei Zhou, Agata Lapedriza, Antonio Torralba, and Aude Oliva. Learning deep features for scene recognition using Places database. In *NeurIPS*, 2014.
- [72] Bolei Zhou, Hang Zhao, Xavier Puig, Tete Xiao, Sanja Fidler, Adela Barriuso, and Antonio Torralba. Semantic understanding of scenes through the ADE20K dataset. *IJCV*, 2019.

A. Implementation Details

A.1. ImageNet Experiments

ViT architecture. We follow the standard ViT architecture [16]. It has a stack of Transformer blocks [57], and each block consists of a multi-head self-attention block and an MLP block, both having LayerNorm (LN) [1]. The encoder ends with LN. As the MAE encoder and decoder have different width, we adopt a linear projection layer after the encoder to match it. Our MAE adds positional embeddings [57] (the sine-cosine version) to both the encoder and decoder inputs. Our MAE does *not* use relative position or layer scaling (which are used in the code of [2]).

We extract features from the encoder output for fine-tuning and linear probing. As ViT has a class token [16], to adapt to this design, in our MAE pre-training we append an auxiliary dummy token to the encoder input. This token will be treated as the class token for training the classifier in linear probing and fine-tuning. Our MAE works similarly well without this token (with average pooling).

Pre-training. The default setting is in Table 8. We do *not* use color jittering, drop path, or gradient clip. We use xavier_uniform [18] to initialize all Transformer blocks, following ViT’s official code [16]. We use the linear lr scaling rule [20]: $lr = base_lr \times batchsize / 256$.

End-to-end fine-tuning. Our fine-tuning follows common practice of supervised ViT training. The default setting is in Table 9. We use layer-wise lr decay [10] following [2].

Linear probing. Our linear classifier training follows [9]. See Table 10. We observe that linear probing requires a very different recipe than end-to-end fine-tuning. In particular, regularization is in general harmful for linear probing. Following [9], we disable many common regularization strategies: we do *not* use mixup [69], cutmix [68], drop path [30], or color jittering, and we set weight decay as zero.

It is a common practice to normalize the classifier input when training a classical linear classifier (*e.g.*, SVM [11]). Similarly, it is beneficial to normalize the pre-trained features when training the linear probing classifier. Following [15], we adopt an extra BatchNorm layer [31] without affine transformation (`affine=False`). This layer is applied on the pre-trained features produced by the encoder, and is before the linear classifier. We note that the layer does *not* break the linear property, and it can be absorbed into the linear classifier after training: it is essentially a re-parameterized linear classifier.³ Introducing this layer helps calibrate the feature magnitudes across different variants in our ablations, so that they can use the same setting without further lr search.

³Alternatively, we can pre-compute the mean and std of the features and use the normalized features to train linear classifiers.

config	value
optimizer	AdamW [39]
base learning rate	1.5e-4
weight decay	0.05
optimizer momentum	$\beta_1, \beta_2=0.9, 0.95$ [6]
batch size	4096
learning rate schedule	cosine decay [38]
warmup epochs [20]	40
augmentation	RandomResizedCrop

Table 8. **Pre-training setting.**

config	value
optimizer	AdamW
base learning rate	1e-3
weight decay	0.05
optimizer momentum	$\beta_1, \beta_2=0.9, 0.999$
layer-wise lr decay [10, 2]	0.75
batch size	1024
learning rate schedule	cosine decay
warmup epochs	5
training epochs	100 (B), 50 (L/H)
augmentation	RandAug (9, 0.5) [12]
label smoothing [52]	0.1
mixup [69]	0.8
cutmix [68]	1.0
drop path [30]	0.1 (B/L) 0.2 (H)

Table 9. **End-to-end fine-tuning setting.**

config	value
optimizer	LARS [66]
base learning rate	0.1
weight decay	0
optimizer momentum	0.9
batch size	16384
learning rate schedule	cosine decay
warmup epochs	10
training epochs	90
augmentation	RandomResizedCrop

Table 10. **Linear probing setting.** We use LARS with a large batch for faster training; SGD works similarly with a 4096 batch.

Partial fine-tuning. Our MAE partial fine-tuning (§4.3) follows the setting in Table 9, except that we adjust the number of fine-tuning epochs. We observe that tuning fewer blocks requires a longer schedule. We set the numbers of fine-tuning epochs as {50, 100, 200} and use the optimal one for each number of blocks tuned.

A.2. Supervised Training ViT-L/H from Scratch

We find that it is nontrivial to train *supervised* ViT-L/H *from scratch* on ImageNet-1K. The training is unstable. While there have been strong baselines with publicly available implementations [53] for smaller models, the recipes for the larger ViT-L/H are unexplored. Directly applying the previous recipes to these larger models does not work. A NaN loss is frequently observed during training.

We provide our recipe in Table 11. We use a wd of 0.3, a large batch size of 4096, and a long warmup, following the original ViT [16]. We use $\beta_2=0.95$ following [6]. We use the regularizations listed in Table 11 and disable others, following [64]. All these choices are for improving training stability. Our recipe can finish training with no NaN loss.

A. 实现细节

A.1. ImageNet 实验

ViT 架构。我们遵循标准 ViT 架构[16]。它由堆叠的 Transformer 块 [57], 组成，每个块包含多头自注意力模块和 MLP 模块，两者都有 LayerNorm (LN) [1]。编码器以 LN 结束。由于 MAE 编码器与解码器宽度不同，我们在编码器之后加入线性投影层以匹配维度。MAE 在编码器和解码器输入中都加入位置嵌入 [57]（正弦-余弦版本）。我们不使用相对位置或层缩放（[2]的代码中使用了这些设计）。

微调与线性探测的特征取自编码器输出。由于 ViT 使用类别令牌 [16],, 为适应该设计，MAE 预训练在编码器输入中追加一个辅助占位令牌。线性探测与微调训练分类器时，将其视为类别令牌。不使用该令牌而采用平均池化，MAE 也同样表现良好。

预训练。默认设置见表 8。不使用颜色扰动、随机深度或梯度裁剪。遵循 ViT 官方代码 [16], 使用 Xavier 均匀初始化 [18] 初始化所有 Transformer 块。采用线性学习率缩放规则 [20]: $lr = \text{基础} lr \times \text{批量大小} / 256$ 。

端到端微调。遵循监督 ViT 训练的常见做法，默认设置见表 9。按 [2]采用逐层学习率衰减 [10]。

线性探测。线性分类器训练遵循 [9], 见表 10。我们发现，线性探测需要与端到端微调很不同的训练方案，尤其是正则化通常有害。遵循 [9], , 关闭多种常用正则化策略：不使用 mixup [69]、cutmix [68]、随机深度 [30],或颜色扰动，并将权重衰减设为零。

训练传统线性分类器（如 SVM [11]）时，归一化分类器输入是一种常见做法。同样，训练线性探测分类器时，归一化预训练特征也有益。遵循 [15], , 我们额外加入不带仿射变换（`affine=False`）的 BatchNorm 层 [31]。该层作用于编码器生成的预训练特征，并位于线性分类器之前。它不会破坏线性性质，训练后可被吸收到线性分类器中：本质上是重新参数化的线性分类器。³ 引入这一层有助于校准不同消融变体的特征幅值，使它们无需进一步搜索学习率就能使用相同设置。

³另一选择是预先计算特征均值与标准差，再用归一化特征训练线性分类器。

配置	值
优化器	AdamW [39]
基础学习率	1.5e-4
权重衰减	0.05
优化器动量	$\beta_1, \beta_2=0.9, 0.95$ [6]
批量大小	4096
学习率日程	余弦衰减 [38]
预热轮数 [20]	40
数据增强	RandomResizedCrop

表 8. 预训练设置。

配置	值
优化器	AdamW
基础学习率	1e-3
权重衰减	0.05
优化器动量	$\beta_1, \beta_2=0.9, 0.999$
逐层学习率衰减 [10, 2]	0.75
批量大小	1024
学习率日程	余弦衰减
预热轮数	5
训练轮数	100 (B), 50 (L/H)
数据增强	RandAug (9, 0.5) [12]
标签平滑 [52]	0.1
mixup [69]	0.8
cutmix [68]	1.0
随机深度 [30]	0.1 (B/L) 0.2 (H)

表 9. 端到端微调设置。

配置	值
优化器	LARS [66]
基础学习率	0.1
权重衰减	0
优化器动量	0.9
批量大小	16384
学习率日程	余弦衰减
预热轮数	10
训练轮数	90
数据增强	RandomResizedCrop

表 10. 线性探测设置。采用大批量 LARS 加速训练；批量 4096 时 SGD 表现相近。

部分微调。MAE 的部分微调（第 4.3 节）遵循表 9 设置，但调整微调轮数。我们观察到，微调的块越少，所需训练日程越长。将微调轮数设为 {50, 100, 200}，并对每个微调块数选择最优值。

A.2. 从零监督训练 ViT-L/H

我们发现，在 ImageNet-1K 上从零监督训练 ViT-L/H 并不容易，训练会不稳定。尽管较小模型已有较强基线及公开实现 [53]，大型 ViT-L/H 的训练方案尚未充分探索。直接将以前的方案用于这些大模型并不可行，训练中经常出现 NaN 损失。

表 11 给出训练方案。遵循原始 ViT [16], 采用 0.3 的权重衰减、4096 的大批量和较长预热；遵循 [6] 使用 $\beta_2=0.95$ 。按 [64]使用表 11 中列出的正则化，并关闭其他策略。所有选择都旨在改善训练稳定性，方案能够完成训练而不出现 NaN 损失。

config	value
optimizer	AdamW
base learning rate	1e-4
weight decay	0.3
optimizer momentum	$\beta_1, \beta_2=0.9, 0.95$
batch size	4096
learning rate schedule	cosine decay
warmup epochs	20
training epochs	300 (B), 200 (L/H)
augmentation	RandAug (9, 0.5) [12]
label smoothing [52]	0.1
mixup [69]	0.8
cutmix [68]	1.0
drop path [30]	0.1 (B), 0.2 (L/H)
exp. moving average (EMA)	0.9999

Table 11. **Supervised training ViT from scratch.**

The accuracy is 82.6% for ViT-L (81.5% w/o EMA), and 83.1% for ViT-H (80.9% w/o EMA). Both ViT-L and ViT-H show an overfitting trend if not using EMA.

As a by-product, our recipe for ViT-B has 82.3% accuracy (82.1% w/o EMA), *vs.* 81.8% in [53].

A.3. Object Detection and Segmentation in COCO

We adapt the vanilla ViT for the use of an FPN backbone [36] in Mask R-CNN [24]. ViT has a stack of Transformer blocks that all produce feature maps at a single scale (*e.g.*, stride 16). We equally divide this stack into 4 subsets and apply convolutions to upsample or downsample the intermediate feature maps for producing different scales (stride 4, 8, 16, or 32, the same as a standard ResNet [25]). FPN is built on these multi-scale maps.

For fair comparisons among different methods, we search for hyper-parameters for each entry in Table 4 (including all competitors). The hyper-parameters we search for are the learning rate, weight decay, drop path rate, and fine-tuning epochs. We will release code along with the specific configurations. For full model and training details, plus additional experiments, see [35].

A.4. Semantic Segmentation in ADE20K

We use UperNet [63] following the semantic segmentation code of [2]. We fine-tune end-to-end for 100 epochs with a batch size of 16. We search for the optimal *lr* for each entry in Table 5 (including all competitors).

The semantic segmentation code of [2] uses relative position bias [49]. Our MAE pre-training does *not* use it. For fair comparison, we turn on relative position bias *only* during transfer learning, initialized as zero. We note that our BEiT reproduction uses relative position bias in *both* pre-training and fine-tuning, following their code.

A.5. Additional Classification Tasks

We follow the setting in Table 9 for iNaturalist and Places fine-tuning (Table 6). We adjust the *lr* and fine-tuning epochs for each individual dataset.

method	model	params	acc
iGPT [6]	iGPT-L	1362 M	69.0
iGPT [6]	iGPT-XL	6801 M	72.0
BEiT [2]	ViT-L	304 M	52.1 [†]
MAE	ViT-B	86 M	68.0
MAE	ViT-L	304 M	75.8
MAE	ViT-H	632 M	76.6

Table 12. **Linear probing results of masked encoding methods.** Our fine-tuning results are in Table 3. [†]: our implementation.

dataset	ViT-B	ViT-L	ViT-H	ViT-H ₄₄₈	prev best
IN-Corruption ↓ [27]	51.7	41.8	33.8	36.8	42.5 [32]
IN-Adversarial [28]	35.9	57.1	68.2	76.7	35.8 [41]
IN-Rendition [26]	48.3	59.9	64.4	66.5	48.7 [41]
IN-Sketch [60]	34.5	45.3	49.6	50.9	36.0 [41]

our supervised training baselines:

IN-Corruption ↓	45.8	42.3	41.3
IN-Adversarial	27.2	29.6	33.1
IN-Rendition	49.4	50.9	50.3
IN-Sketch	35.6	37.5	38.0

Table 13. **Robustness evaluation on ImageNet variants** (top-1 accuracy, except for IN-C [27] which evaluates mean corruption error). We test the same MAE models (Table 3) on different ImageNet validation sets, *without* any specialized fine-tuning. We provide system-level comparisons with the previous best results.

B. Comparison on Linear Probing Results

In §4.3 we have shown that linear probing accuracy and fine-tuning accuracy are largely *uncorrelated* and they have different focuses about linear separability. We notice that existing masked image encoding methods are generally less competitive in linear probing (*e.g.*, than contrastive learning). For completeness, in Table 12 we compare on linear probing accuracy with masking-based methods.

Our MAE with ViT-L has 75.8% linear probing accuracy. This is substantially better than previous masking-based methods. On the other hand, it still lags behind contrastive methods under this protocol: *e.g.*, MoCo v3 [9] has 77.6% linear probing accuracy for the ViT-L (Figure 9).

C. Robustness Evaluation on ImageNet

In Table 13 we evaluate the robustness of our models on different variants of ImageNet validation sets. We use the same models fine-tuned on *original* ImageNet (Table 3) and only run inference on the different validation sets, *without* any specialized fine-tuning. Table 13 shows that our method has strong scaling behavior: increasing the model sizes has significant gains. Increasing the image size helps in all sets but IN-C. Our results outperform the previous best results (of specialized systems) by large margins.

In contrast, *supervised* training performs much worse (Table 13 bottom; models described in A.2). For example, with ViT-H, our MAE pre-training is 35% better on IN-A (68.2% vs 33.1%) than the supervised counterpart.

配置	值
优化器	AdamW
基础学习率	1e-4
权重衰减	0.3
优化器动量	$\beta_1, \beta_2=0.9, 0.95$
批量大小	4096
学习率日程	余弦衰减
预热轮数	20
训练轮数	300 (B), 200 (L/H)
数据增强	RandAug (9, 0.5) [12]
标签平滑 [52]	0.1
mixup [69]	0.8
cutmix [68]	1.0
随机深度 [30]	0.1 (B), 0.2 (L/H)
指数移动平均 (EMA)	0.9999

表 11. 从零监督训练 ViT。

ViT-L 的准确率为 82.6%（不使用 EMA 为 81.5%），ViT-H 为 83.1%（不使用 EMA 为 80.9%）。若不使用 EMA，二者都出现过拟合趋势。

作为附带结果，ViT-B 方案准确率为 82.3%（无 EMA 为 82.1%），而 [53]为 81.8%。

A.3. COCO 目标检测与分割

我们改造普通 ViT，使其作为 Mask R-CNN [24]中的 FPN 骨干[36]。ViT 的 Transformer 块堆叠均生成单一尺度特征图（如步幅 16）。我们将该堆叠均分为 4 个子集，使用卷积对中间特征图上采样或下采样，产生不同尺度（步幅 4、8、16 或 32，与标准 ResNet [25]相同）。FPN 建立在这些多尺度特征图之上。

为公平比较不同方法，我们为表 4 的每个条目（包括所有竞争方法）搜索超参数：学习率、权重衰减、随机深度率及微调轮数。我们会随代码发布具体配置。完整模型、训练细节及额外实验见 [35]。

A.4. ADE20K 语义分割

遵循 [2]的语义分割代码，我们使用 UperNet [63]，以批量大小 16 端到端微调 100 轮，并为表 5 每个条目（包括所有竞争方法）搜索最优学习率。

[2] 的语义分割代码使用相对位置偏置 [49]，而 MAE 预训练不使用。为公平比较，仅在迁移学习时启用该偏置，并初始化为零。我们复现 BEiT 时遵循其代码，在预训练和微调中都使用相对位置偏置。

A.5. 其他分类任务

iNaturalist 与 Places 微调（表 6）遵循表 9。对每个数据集分别调整学习率与微调轮数。

方法	模型	参数量	acc
iGPT [6]	iGPT-L	1362 M	69.0
iGPT [6]	iGPT-XL	6801 M	72.0
BEiT [2]	ViT-L	304 M	52.1 [†]
MAE	ViT-B	86 M	68.0
MAE	ViT-L	304 M	75.8
MAE	ViT-H	632 M	76.6

表 12. 掩码编码方法的线性探测结果。我们的微调结果见表 3。[†]: 我们的实现。

数据集	ViT-B	ViT-L	ViT-H	ViT-H ₄₄₈	此前最佳
IN-Corruption ↓ [27]	51.7	41.8	33.8	36.8	42.5 [32]
IN-Adversarial [28]	35.9	57.1	68.2	76.7	35.8 [41]
IN-Rendition [26]	48.3	59.9	64.4	66.5	48.7 [41]
IN-Sketch [60]	34.5	45.3	49.6	50.9	36.0 [41]

我们的监督训练基线:

IN-Corruption ↓	45.8	42.3	41.3
IN-Adversarial	27.2	29.6	33.1
IN-Rendition	49.4	50.9	50.3
IN-Sketch	35.6	37.5	38.0

表 13. ImageNet 变体上的鲁棒性评估（top-1 准确率；IN-C [27] 评估平均破坏误差）。在不同 ImageNet 验证集测试相同 MAE 模型（表 3），不做针对性微调，与此前最佳系统结果比较。

B. 线性探测结果比较

第 4.3 节已表明，线性探测与微调准确率在很大程度上不相关，它们对线性可分性的关注不同。现有掩码图像编码方法在线性探测中通常较缺乏竞争力（例如相比对比学习）。为完整起见，表 12 比较了基于遮蔽的方法的线性探测准确率。

ViT-L 的 MAE 线性探测准确率为 75.8%，显著优于此前基于遮蔽的方法。但该协议下仍落后于对比方法，例如 MoCo v3 [9] 的 ViT-L 线性探测准确率为 77.6%（图 9）。

C. ImageNet 鲁棒性评估

表 13 在 ImageNet 验证集的不同变体上评估模型鲁棒性。我们使用在原始 ImageNet 上微调的相同模型（表 3），只在不同验证集上推理，不做针对性微调。表 13 展示了良好的规模扩展行为：增大模型可获得显著收益。提高图像尺寸对除 IN-C 之外的所有集合都有帮助。我们的结果大幅超过此前专用系统的最佳结果。

相比之下，监督训练差很多（表 13 下部；模型见 A.2）。例如使用 ViT-H 时，MAE 预训练在 IN-A 上比监督模型高 35%（68.2% 对 33.1%）。

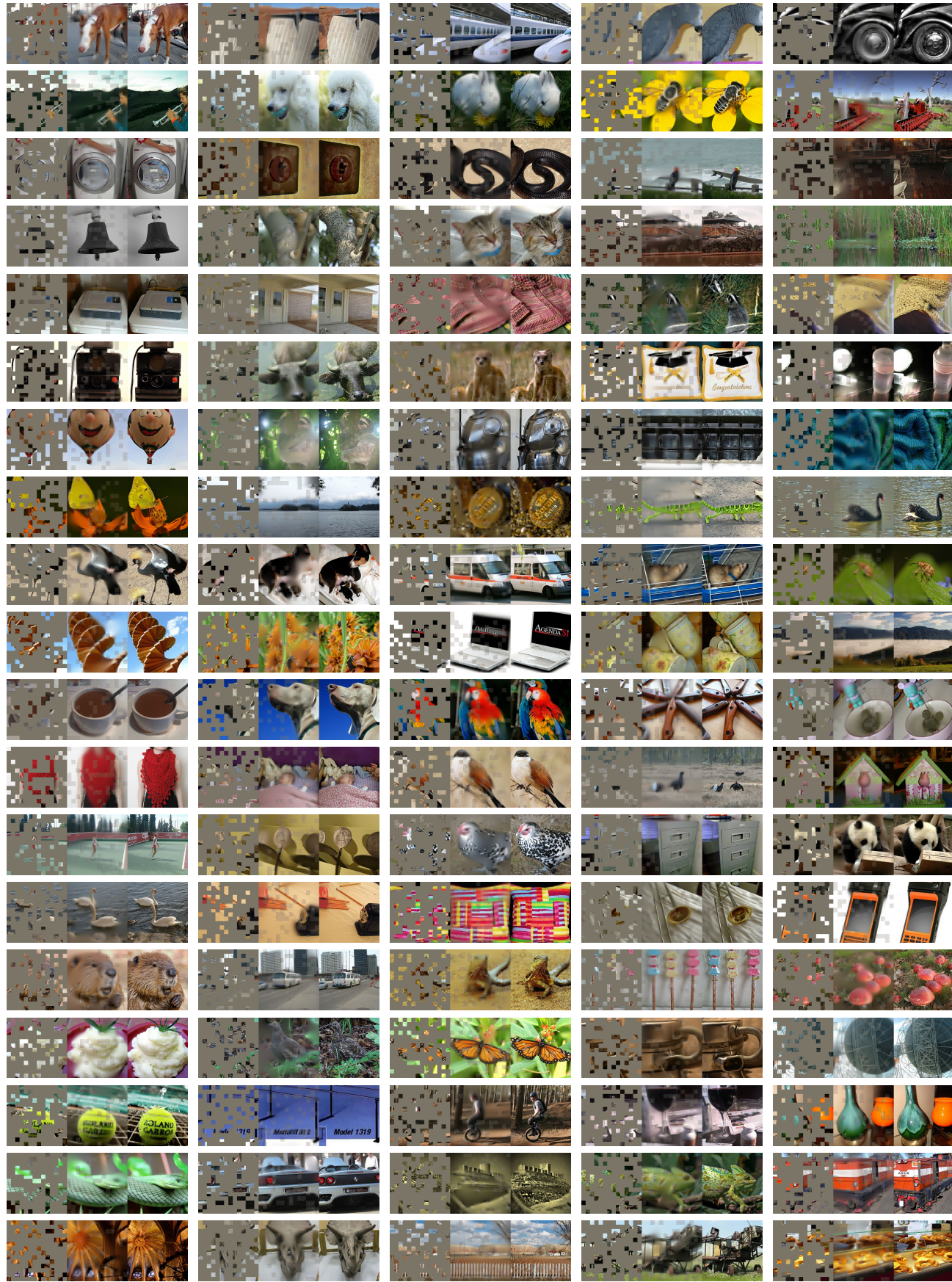


Figure 10. **Uncurated random samples** on ImageNet *validation* images. For each triplet, we show the masked image (left), our MAE reconstruction (middle), and the ground-truth (right). The masking ratio is 75%.

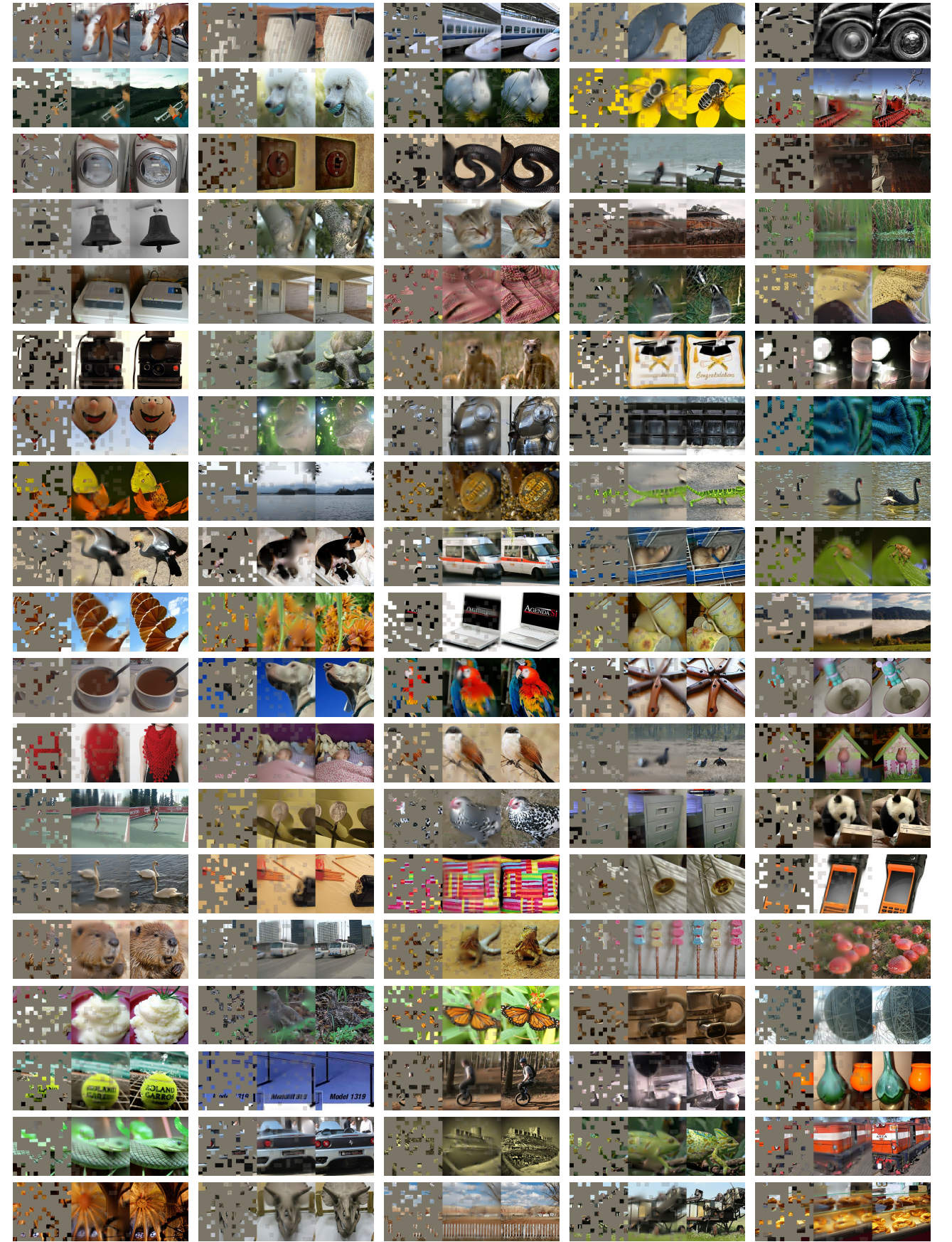


图 10. ImageNet 验证集上未经筛选的随机样本。每组三幅图依次为遮蔽图像（左）、MAE 重建（中）和真实图像（右）。掩码比例 75%。

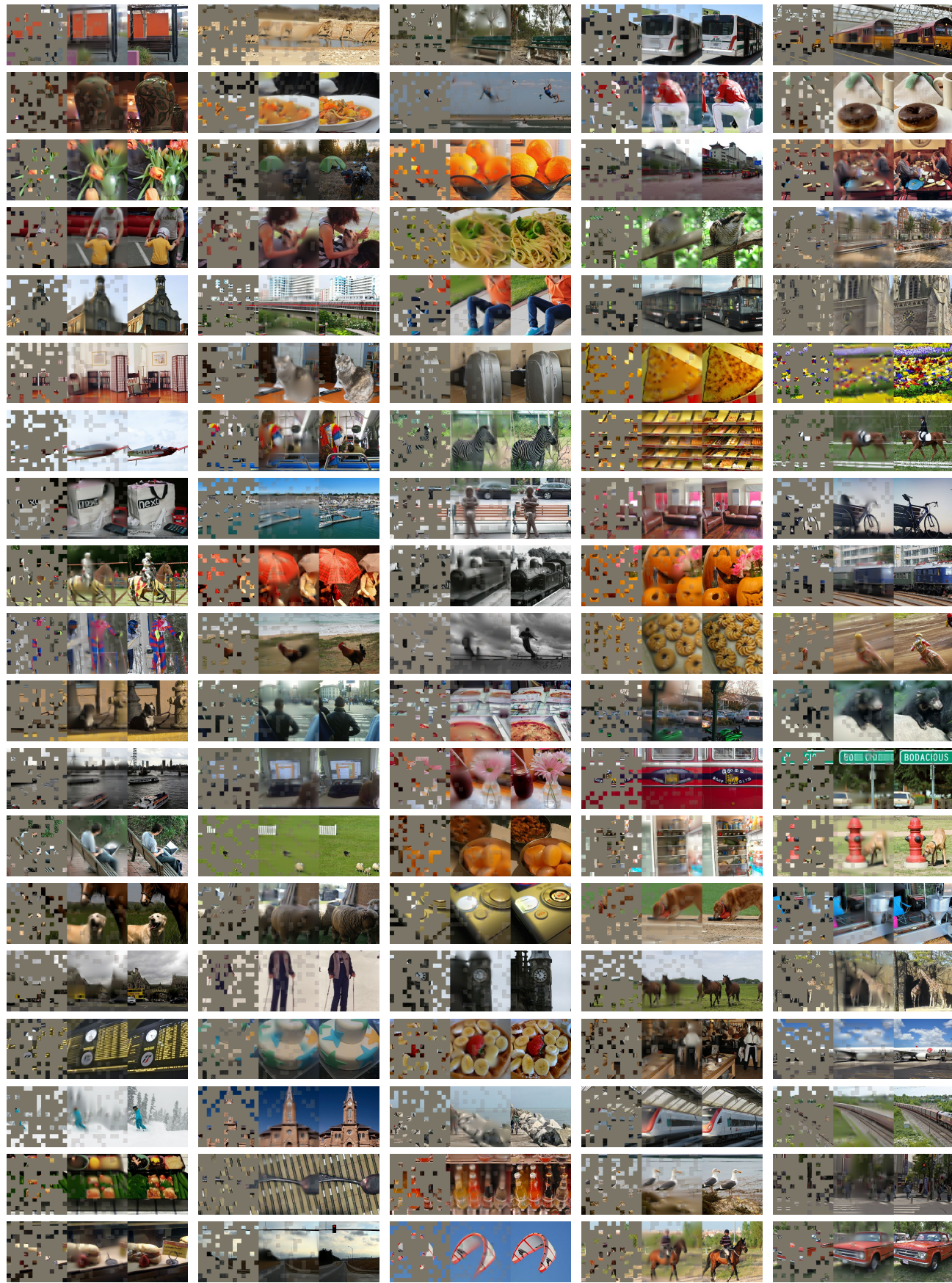


Figure 11. **Uncurated random samples** on COCO validation images, using an MAE trained on ImageNet. For each triplet, we show the masked image (left), our MAE reconstruction (middle), and the ground-truth (right). The masking ratio is 75%.

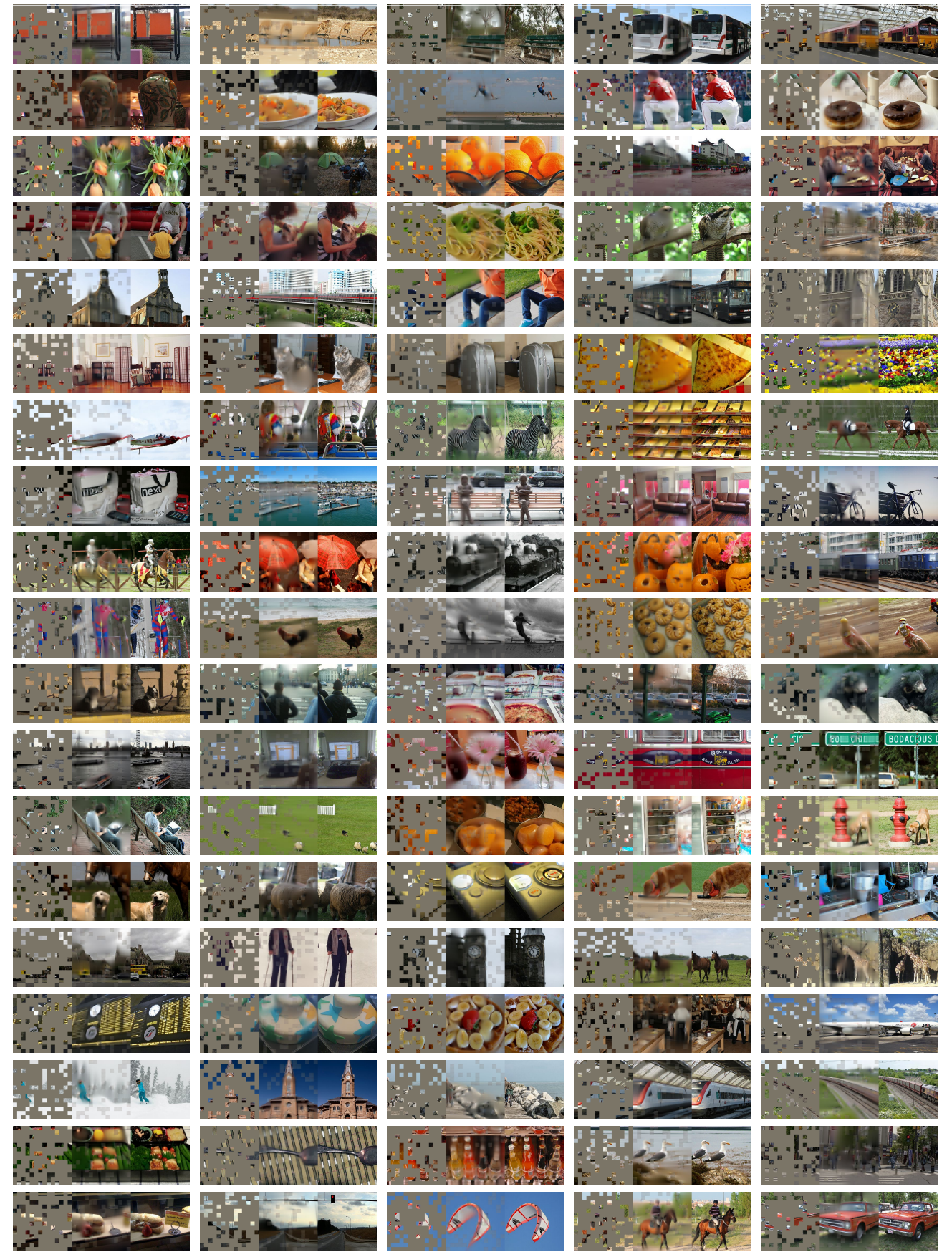


图 11. COCO 验证集上未经筛选的随机样本，使用在 ImageNet 上训练的 MAE。每组三幅图依次为遮蔽图像（左）、MAE 重建（中）和真实图像（右）。掩码比例为 75%。